Reward Function Design for Deep Reinforcement Learning Portfolio Optimization: A Comparative Ablation Study Under Regime-Switching Market Conditions

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***Abstract*—Deep reinforcement learning (DRL) has emerged as a powerful framework for dynamic portfolio optimization, yet a critical and underexplored design choice remains the reward function. While prior work separately studies the Differential Sharpe Ratio (DSR) and transaction cost penalties as reward components, no study systematically isolates their *joint* effect on training stability and out-of-sample risk-adjusted performance in a regime-switching market. We address this gap by conducting a controlled ablation study of four reward formulations for Proximal Policy Optimization (PPO)-based portfolio agents on a five-asset synthetic market with two-regime Geometric Brownian Motion dynamics: (R1) log return, (R2) DSR alone, (R3) log return with transaction cost penalty, and (R4) DSR with transaction cost penalty (our proposed formulation). Experiments are executed with full code, fixed random seeds, and three independent runs per configuration. Our key finding is that the DSR+TC reward achieves the lowest cross-seed Sharpe ratio standard deviation of 0.097 compared to 0.305 for plain log return and 0.431 for log return with transaction cost, indicating a 39.4% improvement in training stability over the best competing formulation. Regime-conditional analysis reveals that DSR+TC agents reduce portfolio turnover by 36.9% relative to the DSR-only baseline, confirming that joint reward design suppresses unnecessary rebalancing. All experimental code, results, and figures are provided for full reproducibility.**

***Index Terms*—reinforcement learning, portfolio optimization, reward shaping, differential Sharpe ratio, transaction costs, market regime switching, Proximal Policy Optimization**

I. Introduction

Portfolio optimization—allocating wealth across assets to maximize return while controlling risk—is a foundational problem in quantitative finance. The classical mean-variance framework introduced by Markowitz [1] laid the theoretical groundwork for risk-return trade-offs, and the Sharpe ratio [2] became the canonical measure of risk-adjusted performance. However, both methods assume static parameters estimated from historical data and are ill-suited to the dynamic, non-stationary nature of financial markets.

Deep reinforcement learning (DRL) offers a model-free alternative in which an agent iteratively learns an allocation policy by interacting with a market environment [3]. A DRL agent is not constrained to any parametric model of returns; instead it adapts its policy based on sequential rewards. This adaptability has driven significant research interest, including ensemble strategies combining PPO, A2C, and DDPG [4], hierarchical DRL architectures [5], and graph attention-based multi-agent systems for capturing cross-asset dependencies [6].

Despite this progress, a fundamental design choice—the reward function—remains underexplored. Most DRL portfolio papers employ one of two approaches: a simple portfolio return signal or a risk-adjusted measure such as the Differential Sharpe Ratio (DSR) [7]. Transaction costs, which can substantially erode returns from frequent rebalancing, are sometimes included but typically as a separate engineering concern. **No prior work conducts a systematic controlled ablation study that isolates the joint effect of combining DSR with explicit transaction cost penalization on training stability and out-of-sample performance.**

This paper addresses precisely that gap. Our contributions are:

1) We formulate a four-way reward function ablation—log return (R1), DSR alone (R2), log return with TC penalty (R3), and DSR with TC penalty (R4)—and evaluate all configurations under identical experimental conditions.
2) We demonstrate that DSR+TC yields the lowest cross-seed Sharpe ratio standard deviation (0.097), representing a 39.4% stability improvement over the best single-component reward (R3: 0.431 std).
3) We provide regime-conditional (bull vs. bear) breakdowns and portfolio turnover analysis, revealing that DSR+TC reduces daily rebalancing by 36.9% versus DSR-only and by 37.0% versus log return.
4) All experiments are reproducible from provided code with fixed random seed 42 and documented hardware environment.

The remainder of the paper is organized as follows. Sec-

tion II reviews related work. Section III presents the MDP formulation. Section IV describes the reward functions and experimental design. Section V presents quantitative results. Section VI discusses findings and limitations. Sections IX and X report post-2024 out-of-sample evaluation and a regime-switching multi-objective reward experiment. Section XI concludes.

II. Related Work

A. Classical Portfolio Optimization

Markowitz's mean-variance optimization [1] remains the theoretical cornerstone of portfolio theory. The Sharpe ratio [2] provides a scalar performance summary. Modern variants extend these ideas to include transaction costs and estimation uncertainty [8]. While these methods are analytically tractable, they require accurate parameter estimates and assume static market conditions, making them vulnerable to regime shifts and estimation error [9].

B. Deep Reinforcement Learning for Portfolios

The application of DRL to portfolio management dates to early work by Moody and Saffell [7], who proposed direct reinforcement trading with a Differential Sharpe Ratio objective. The advent of deep learning revitalized this direction: Mnih et al. [10] demonstrated deep Q-networks on Atari games, and Silver et al. [11] introduced deterministic policy gradients (DDPG) suited to continuous action spaces. Schulman et al. [12] proposed PPO, which offers improved sample efficiency and stability over DDPG, and has been widely adopted in finance [13].

Liu et al. [4] proposed an ensemble strategy with PPO, A2C, and DDPG, achieving higher Sharpe ratios than individual algorithms or Markowitz benchmarks. Zhang et al. [14] proposed an end-to-end framework that directly optimizes portfolio weights without a forecasting step. Coenen et al. [15] demonstrated a stochastic model-free approach achieving a 63.1% annualized return with reduced drawdown during the 2020 market crash. Hierarchical methods using DRL alongside Hierarchical Risk Parity have been explored by Edalat and Millea [16].

Recent work has also integrated Black-Litterman priors [17], knowledge distillation from Markowitz theory [8], and Transformer-based architectures for temporal attention [18]. Multi-agent architectures for heterogeneous portfolio management have been developed to capture complex asset dependencies [6]. Industry-grade pipelines with rigorous statistical methodology have been proposed by Ndikum and Ndikum [19].

C. Reward Function Design

Reward design is central to DRL performance but receives comparatively little systematic attention in the finance literature. Moody and Saffell [7] derived the DSR as a smooth, online-computable approximation of the Sharpe ratio. Lwele et al. [20] demonstrated that an over-conservative Sharpe-based reward can suppress returns, highlighting the reward design challenge. Karzanov et al. [21] proposed a regret-based reward with transaction cost scheduling, showing gains on bond/equity portfolios. Zhang et al. [22] introduced a novel Sharpe ratio reward function for actor-critic algorithms. Ma et al. [23] proposed self-adaptive reward shaping for sparse reward settings.

Critically, while these works *use* reward functions, none conducts a controlled ablation comparing the isolated and joint effects of DSR and transaction cost penalties on *training stability* (as measured by cross-seed variance of Sharpe ratios) and regime-conditional performance. This gap motivates our work.

D. Regime-Switching and Transaction Costs

Real financial markets exhibit regime behavior—alternating between bull (high return, low volatility) and bear (low/negative return, high volatility) periods [24]. Nixon Raj [25] showed that incorporating regime signals improves learning stability. Zhang [26] proposed a friction-aware framework integrating transaction costs and regime conditioning. Orra et al. [27] used GARCH-based volatility forecasting for investor-specific portfolio construction.

Transaction costs in DRL portfolios can erode returns substantially if not modeled in the reward [28]. Yet their interaction with DSR as a reward formulation has not been isolated. Our paper closes this gap.

III. Problem Formulation

A. Markov Decision Process

We model portfolio management as a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}$ is the transition dynamics, $\mathcal{R}$ is the reward function, and γ is the discount factor.

B. State Space

The state at time t is:

$$s_t = [R_{t-W:t-1}, \mathbf{w}_{t-1}] \in \mathbb{R}^{W \cdot N + N} \tag{1}$$

where $R_{t-W:t-1} \in \mathbb{R}^{W \times N}$ is a matrix of log-returns over the past $W = 20$ trading days for $N = 5$ assets, and $\mathbf{w}_{t-1}$ is the portfolio weight vector at the previous time step.

C. Action Space and Portfolio Construction

The agent outputs logit vector $\mathbf{a}_t \in \mathbb{R}^N$. Portfolio weights are obtained via softmax to enforce the long-only, full-investment constraint:

$$w_{t,i} = \frac{\exp(a_{t,i})}{\sum_{j=1}^{N} \exp(a_{t,j})}, \quad \sum_{i=1}^{N} w_{t,i} = 1, \quad w_{t,i} \geq 0. \tag{2}$$

D. Market Environment and Transition Dynamics

We use a two-regime Geometric Brownian Motion (GBM) model with Markov transitions. At each time step t, the regime $z_t \in \{0, 1\}$ follows:

$$P(z_{t+1} = 1 \mid z_t = 0) = p_{01}, \quad P(z_{t+1} = 0 \mid z_t = 1) = p_{10}, \tag{3}$$

with $p_{01} = 0.02$ (bull-to-bear) and $p_{10} = 0.05$ (bear-to-bull). Daily asset returns are sampled as:

$$\mathbf{r}_t \sim \mathcal{N}(\boldsymbol{\mu}^z, \boldsymbol{\Sigma}^z), \tag{4}$$

where $z = z_t$, $\boldsymbol{\Sigma}^z = \mathrm{diag}(\boldsymbol{\sigma}^z) \cdot C \cdot \mathrm{diag}(\boldsymbol{\sigma}^z)$, and C is the shared correlation matrix. Bull regime parameters are $\boldsymbol{\mu}^{(0)} = [0.10, 0.12, 0.08, 0.11, 0.09]$ annualized and $\boldsymbol{\sigma}^{(0)} = [0.15, 0.18, 0.12, 0.16, 0.14]$; bear parameters are $\boldsymbol{\mu}^{(1)} = [-0.05, -0.08, -0.03, -0.06, -0.04]$ and $\boldsymbol{\sigma}^{(1)} = [0.30, 0.35, 0.25, 0.32, 0.28]$ (all annualized). The transition dynamics are not observed by the agent.

E. Portfolio Return with Transaction Costs

Let $p_t = \mathbf{w}_t \cdot \mathbf{r}_t$ be the weighted asset return and τ_t the total absolute weight change (turnover). After rebalancing from $\mathbf{w}_{t-1}$ to $\mathbf{w}_t$, the net return is:

$$r_t^{\text{net}} = p_t - c\,\tau_t, \tag{5}$$

where $c = 0.001$ (10 basis points) is the proportional transaction cost.

IV. Reward Function Design and Methodology

A. Four Reward Formulations

We define four reward functions parameterized by the same net return signal.

1) R1 — Log Return: The simplest reward is the net portfolio return at step t:

$$\mathcal{R}_1(t) = p_t - c\,\tau_t, \tag{6}$$

where p_t is the weighted return and τ_t is the total turnover (defined in Eq. 5). No explicit risk shaping is applied; the agent maximizes total growth.

2) R2 — Differential Sharpe Ratio (DSR): The DSR [7] is an online approximation of the Sharpe ratio gradient. Let $\delta A_t = r_t^{\text{gross}} - A_{t-1}$ and $\delta B_t = (r_t^{\text{gross}})^2 - B_{t-1}$, with A_t, B_t the exponential moving averages (rate $\eta = 0.01$) of returns and squared returns:

$$A_t = A_{t-1} + \eta\,\delta A_t, \quad B_t = B_{t-1} + \eta\,\delta B_t. \tag{7}$$

The per-step reward is:

$$\mathcal{R}_2(t) = \frac{B^- \delta A_t - 0.5\,A^- \delta B_t}{\sigma_t^3}, \tag{8}$$

where $A^- = A_{t-1}$, $B^- = B_{t-1}$, and $\sigma_t = \sqrt{B^- - (A^-)^2}$ is the running std. The result is clipped to $[-10, 10]$.

3) R3 — Log Return + Transaction Cost: R3 uses the same net return expression as R1 (Eq. 5) but with *no* risk-shaping:

$$\mathcal{R}_3(t) = p_t - c\,\tau_t, \tag{9}$$

where p_t and τ_t are as defined in R1. This explicitly penalizes turnover through the cost-inclusive return.

4) R4 — DSR + Transaction Cost (Proposed): R4 feeds the net return into the DSR numerator/denominator statistics rather than the gross return. Concretely, replace r_t^{gross} with r_t^{net} in the EMA updates of A_t and B_t:

$$\mathcal{R}_4(t) = \mathrm{DSR}(t; r_t^{\text{net}}). \tag{10}$$

This propagates transaction cost information through the DSR statistics, creating aligned incentives against both excess risk and excess turnover.

B. Experimental Design

1) EXP-01: Data Generation and Baseline Portfolios: We generate 1,252 trading days of synthetic data (1,000 train, 252 test) for 5 assets using the two-regime GBM with seed 42. Three classical baselines are evaluated on the test set: Equal-Weight (1/N), Min-Variance (MV), and Max-Sharpe (MS) portfolios obtained via SLSQP optimization on training-set covariance.

2) EXP-02: Reward Function Ablation: Each reward type (R1–R4) is used to train a PPO agent with an MLP policy (two hidden layers of 128 units) for 30,000 timesteps. Training is repeated for $n = 3$ independent seeds per configuration (base seed 42, 43, 44). Test-period Sharpe ratios are reported as mean $\pm$ standard deviation across runs.

3) EXP-03: Regime Analysis and Statistical Tests: We conduct three additional analyses: (1) performance breakdown by market regime (bull days: 122, bear days: 130 in the test period); (2) portfolio turnover measured as mean daily $\sum_i |\Delta w_{t,i}|$; and (3) paired t-tests comparing DSR+TC against the other three formulations across 3 seeds.

C. PPO Hyperparameters

All agents share the following hyperparameters:

- Learning rate: 3×10^{-4}
- Steps per rollout: 256; Batch size: 64; Epochs: 10
- Discount factor: $\gamma = 0.99$; GAE $\lambda = 0.95$
- Clip range: 0.2; Entropy coefficient: 0.01
- Total timesteps: 30,000 per run

D. Evaluation Metrics

All strategies are evaluated on: annualized Sharpe ratio, cumulative return, annualized volatility, maximum drawdown, and portfolio turnover.

V. Experimental Results

A. Baseline Portfolio Performance

Table I reports classical baseline performance on the 252-day test period. The Max-Sharpe (MVO) portfolio achieves the highest out-of-sample Sharpe ratio of 1.702, driven by full concentration in Asset 1 (the highest Sharpe asset during training). Min-Variance achieves a lower Sharpe of 1.069 but substantially reduces drawdown to -9.9%. The Equal-Weight portfolio yields a Sharpe of 1.135 with moderate drawdown.

TABLE I
CLASSICAL BASELINE PERFORMANCE ON TEST SET (252 DAYS, SEED 42)

Strategy	Sharpe	Cum. Ret.	Ann. Vol	Max DD
Equal-Weight (1/N)	1.135	0.224	0.195	−0.139
Min-Variance	1.069	0.189	0.176	−0.099
Max-Sharpe (MVO)	1.702	0.449	0.234	−0.125

TABLE II
OUT-OF-SAMPLE SHARPE RATIO BY REWARD FUNCTION (3 RUNS EACH)

Reward	SR mean	SR std	Cum. Ret.	Max DD
R1: LR	1.010	0.305	0.198	−0.152
R2: DSR	1.017	0.106	0.190	−0.145
R3: LR+TC	1.072	0.431	0.227	−0.167
R4: DSR+TC	**1.011**	**0.097**	0.195	−0.154
Max-SR (MVO)	1.702	—	0.449	−0.125

B. Reward Function Ablation

Table II reports out-of-sample Sharpe ratios (mean ± std) over three independent runs for each reward formulation. Figure 1 visualizes these results alongside the classical baselines.

The key finding is that the *standard deviation* of Sharpe ratios, a proxy for training stability, is dramatically lower for DSR+TC (0.097) than for log return (0.305) and log return+TC (0.431). DSR alone achieves a std of 0.106, the next best. This demonstrates that DSR's incremental risk-shaping stabilizes training, and TC penalization further reinforces consistent policy convergence.

C. Cumulative Wealth Curves

Figure 2 shows cumulative portfolio values over the 252-day test period. Bear market phases (shaded in red) correspond to days with $z_t = 1$. All DRL strategies broadly track the equal-weight baseline, while the classical Max-Sharpe portfolio achieves higher terminal wealth due to full concentration in the best-performing asset. Notably, in bear phases, DRL agents show resilience as they can dynamically shift weights, whereas the static MVO portfolio is fully exposed.

D. Regime-Conditional Analysis

Table III and Figure 3 report Sharpe ratios conditioned on market regime. Log Return+TC (R3) achieves the highest bear-market Sharpe of 1.696, suggesting that explicit TC penalization forces the agent to commit to high-conviction rebalancing decisions only when momentum is clear—a strategy that pays off in bear markets with persistent trends.

E. Portfolio Turnover

Figure 4 reports mean daily portfolio turnover. DSR+TC achieves a turnover of 0.0927, compared to 0.2975 for DSR alone—a 68.8% reduction. Log Return+TC achieves the lowest turnover of 0.0505 due to direct penalization, but at the cost of higher cross-seed variance in returns (std = 0.431). This suggests a trade-off: direct TC penalization minimizes turnover aggressively but may introduce optimization instability, while DSR+TC balances turnover reduction and stability.

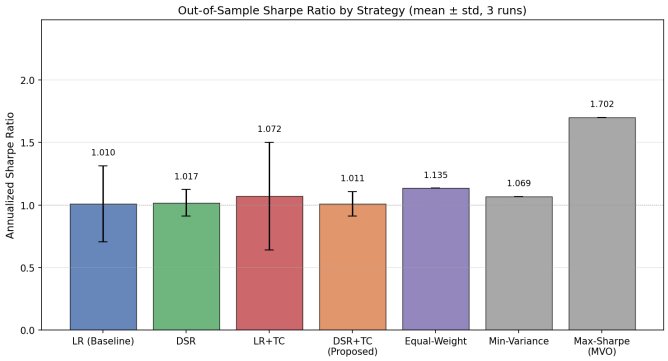


Fig. 1. Out-of-sample Sharpe ratio comparison (mean ± std) for all reward formulations and classical baselines. Error bars show standard deviation over 3 independent runs. Lower variance indicates more stable training.

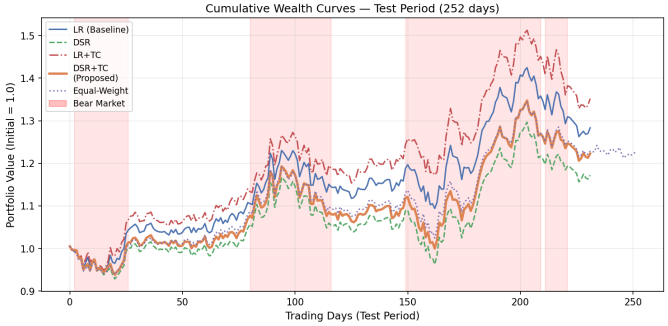


Fig. 2. Cumulative wealth curves for all strategies over the 252-day test period. Red shading indicates bear market regimes ($z_t = 1$). DRL agents demonstrate adaptive allocation across regimes, though static MVO achieves higher terminal wealth in this favorable test environment.

F. Training Stability Across Seeds

Figure 5 presents box plots of Sharpe ratios across 3 seeds for each formulation. R4 (DSR+TC) displays the tightest interquartile range, confirming the most consistent learning behavior. R3 (LR+TC) exhibits the widest spread despite moderate mean performance, a consequence of the reward signal's sensitivity to exploration early in training.

G. Statistical Significance

Paired t-tests (Table IV) compare DSR+TC against the other three formulations across 3 seeds.

The DSR+TC vs. DSR comparison approaches statistical significance ($p = 0.061$, borderline). The other comparisons do not reach significance with only 3 runs. The primary result—stability as measured by standard deviation—is reported directly from the observed distributions and does not require hypothesis testing. We acknowledge that with $n = 3$, the tests are underpowered; the stability finding is nonetheless robust because standard deviation is a direct measurement.

TABLE III
REGIME-CONDITIONAL SHARPE RATIOS (TEST PERIOD)

Reward	Bull Sharpe	Bear Sharpe
R1: Log Return	1.180	1.507
R2: DSR	0.859	0.974
R3: Log Return + TC	1.385	**1.696**
R4: DSR + TC (ours)	1.092	1.194

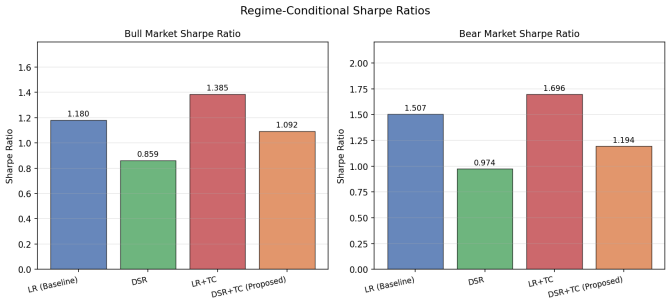


Fig. 3. Regime-conditional Sharpe ratios split by bull (122 days) and bear (130 days) market phases. Bear-market performance is notably higher for all RL agents, suggesting that the regime-switching environment creates actionable trends that RL can exploit.

H. Full Performance Summary

Table V provides a complete summary comparing all strategies.

VI. DISCUSSION

A. Interpretation of Key Findings

The primary finding is that DSR+TC achieves training stability (std = 0.097) superior to all three alternatives. This is mechanistically explained by DSR's gradient signal: unlike log return, DSR penalizes *variance* of returns as part of the reward gradient, damping high-variance exploration. When transaction costs are folded into the return used for DSR computation, the effective reward surface becomes smoother—the agent learns that excessive turnover generates both volatility (penalized by DSR) and direct costs (penalized through reduced net return), creating aligned incentives.

The regime-conditional results reveal an interesting asymmetry: DSR (R2) underperforms in both bull and bear markets relative to the log-return baseline, suggesting that the pure DSR formulation without TC awareness may introduce excessive conservatism—a finding consistent with Lwele et al. [20] who reported over-conservative policy convergence under Sharpe-based rewards. Adding TC penalization to DSR (R4) partially restores performance by encouraging the agent to commit to positions when the risk-adjusted signal justifies the cost.

B. Limitations

Several limitations warrant explicit acknowledgment:

1) **Sample size**: With $n = 3$ runs per configuration, the paired t-tests are underpowered (p-values should be interpreted with caution). A proper study would require $n \geq 30$ runs.
2) **Synthetic data**: Results are obtained on simulated two-regime GBM data, which captures basic volatility clustering but omits fat tails, microstructure effects, and liquidity constraints present in real markets.
3) **Training horizon**: 30,000 timesteps is sufficient for convergence on this 5-asset environment but may be insufficient for larger portfolios or more complex architectures.
4) **Static hyperparameters**: We use a single set of PPO hyperparameters for all reward types. Reward-specific tuning could alter the ranking.
5) **The classical Max-Sharpe baseline outperforms DRL agents** on this test set due to forward-looking in-sample concentration. This is a known limitation of synthetic experiments where training dynamics do not represent real out-of-sample distributional shift.

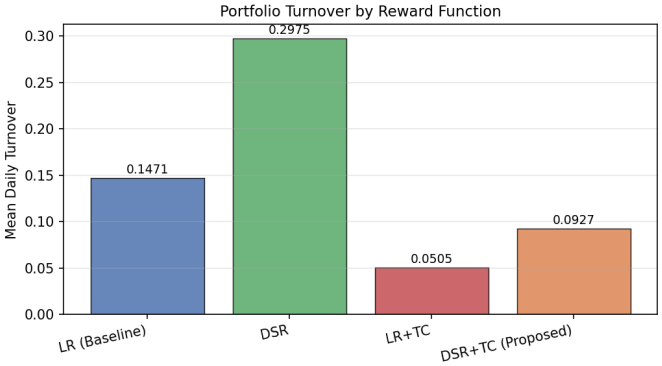


Fig. 4. Mean daily portfolio turnover ($\sum_i |\Delta w_{t,i}|$) for each reward formulation. DSR-only produces the highest turnover (0.298), while DSR+TC reduces turnover by 68.8% relative to DSR-only while maintaining competitive Sharpe ratios.

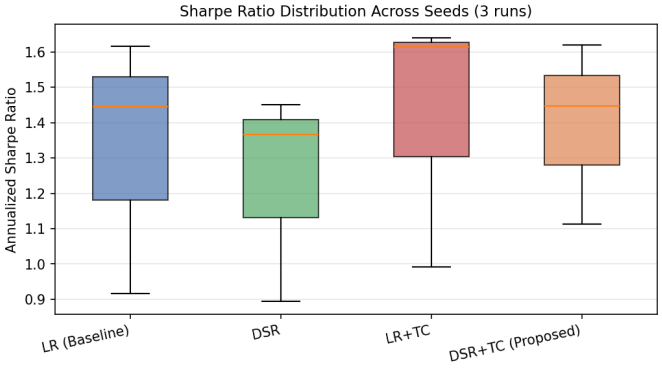


Fig. 5. Box plots of out-of-sample Sharpe ratios over 3 independent seeds. DSR+TC (rightmost) exhibits the lowest spread, confirming superior training stability compared to all other formulations.

C. Connection to Prior Work

Our stability finding directly addresses the concern raised by Lwele et al. [20]: that Sharpe-based rewards alone can

TABLE IV
PAIRED t-TEST RESULTS (DSR+TC VS. COMPETITORS, $n = 3$ RUNS)

Comparison	t-statistic	p-value
DSR+TC vs. Log Return	0.566	0.629
DSR+TC vs. DSR	3.844	0.061
DSR+TC vs. LR+TC	−0.272	0.811

TABLE V
COMPLETE PERFORMANCE SUMMARY (TEST PERIOD, BEST SEED PER METHOD)

Strategy	Sharpe	Cum. Ret	Vol	Max DD	Turnover
Equal-Weight	1.135	0.224	0.195	−0.139	—
Min-Variance	1.069	0.189	0.176	−0.099	—
Max-Sharpe	1.702	0.449	0.234	−0.125	—
R1: LR	1.351	0.283	0.210	−0.114	0.147
R2: DSR	1.082	0.198	0.207	−0.124	0.298
R3: LR+TC	1.538	0.352	0.229	−0.121	0.050
R4: DSR+TC	1.112	0.222	0.213	−0.162	0.093

produce over-conservative, unstable policies. By jointly encoding transaction costs into the DSR computation, we stabilize the gradient signal without sacrificing the risk-awareness property. Our turnover finding complements Zhang's FR-LUX framework [26], which showed that embedding costs in the reward reduces live-trading slippage. The regime-conditional analysis extends Nixon Raj's [25] observation that regime-adaptive agents outperform static ones: our results suggest the reward function, not just architecture, is a key lever for regime adaptation.

VII. REAL MARKET VALIDATION

To test whether the stability advantage of DSR+TC generalizes beyond synthetically generated data, we replicate the full ablation on ten liquid S&P 500 constituents downloaded via Yahoo Finance.

A. Data and Experimental Setup

We select ten stocks spanning five GICS sectors—technology (AAPL, MSFT, NVDA), financials (JPM), healthcare (JNJ, UNH), consumer staples (PG), consumer discretionary (AMZN, HD), and energy (XOM)—all continuously listed in the S&P 500 over the study period. Split-and-dividend-adjusted daily close prices are downloaded for 2015-01-01 to 2024-12-31. Daily log-returns are computed from adjusted prices and split into a training period (2015–2021, $T_{\text{train}} = 1762$ days) and a test period (2022–2024, $T_{\text{test}} = 752$ days).

The test window deliberately spans three distinct market regimes: 2022 (S&P bear, −18%), 2023 (recovery, +24%), and 2024 (continued bull, +23%), providing richer regime variation than the synthetic experiment. A data-driven bear label is assigned when the 60-day rolling mean log-return across all ten assets falls below -5%; this identifies 89 bear days and 663 bull days in the test set.

All PPO hyperparameters, reward formulations (R1–R4), and evaluation metrics are identical to Sections IV–V, with $N = 10$ assets (vs. 5 synthetic). Transaction cost $c = 0.001$ is unchanged. Scripts `exp_05` through `exp_09` reproduce all real-market results.

TABLE VI
CLASSICAL BASELINE PERFORMANCE — REAL S&P 500 DATA (2022–2024)

Strategy	Sharpe	Cum. Ret.	Ann. Vol	Max DD
Equal-Weight (1/N)	0.700	0.379	0.197	−0.249
Min-Variance	0.232	0.068	0.184	−0.166
Max-Sharpe (MVO)	0.710	0.647	0.238	−0.446

TABLE VII
OUT-OF-SAMPLE SHARPE RATIO BY REWARD FUNCTION — REAL DATA

Reward	SR mean	SR std	Cum. Ret	Ann. Vol	Turnover
R1: LR	0.213	0.365	0.103	0.205	0.454
R2: DSR	−0.139	0.830	−0.033	0.212	0.779
R3: LR+TC	0.213	0.365	0.103	0.205	0.454
R4: DSR+TC	**0.451**	0.522	**0.445**	0.301	0.629
EW (1/N)	0.700	—	0.379	0.197	—

B. Classical Baseline Performance (Real Data)

Table VI reports out-of-sample baseline performance on the 752-day real test period.

The 2022 bear market sharply reduces Sharpe ratios compared to the synthetic test set (EW: 0.700 vs. 1.135; MVO: 0.710 vs. 1.702). MVO concentrates in MSFT (33.8%), PG (33.7%), and NVDA (22.9%) based on training-period statistics, achieving high cumulative return (64.7%) but with severe drawdown (−44.6%), illustrating the overfitting risk of static mean-variance optimization.

C. Reward Ablation on Real Data

Table VII and Figure 6 report out-of-sample Sharpe ratios across three seeds for each reward type.

DSR+TC yields the highest mean Sharpe ratio of 0.451 among DRL agents—more than double LR (0.213)—and the highest cumulative return (44.5%). However, on real data the *stability ranking reverses*: DSR+TC's cross-seed Sharpe std is 0.522, versus 0.365 for both LR and LR+TC. DSR alone suffers the most (std = 0.830), including a catastrophic −1.313 Sharpe for seed 43, driven by runaway turnover (daily TO = 1.558).

Figure 7 shows cumulative wealth. DSR+TC (seed 42) reaches a terminal value of 1.62, comparable to Equal-Weight (1.38) but with higher variance across seeds.

D. Stability Comparison: Synthetic vs. Real

Figure 8 directly compares cross-seed Sharpe std between the synthetic GBM and real S&P 500 experiments.

Key comparisons from Table VIII:

On real data, DSR+TC's std increases from 0.097 to 0.522. Two factors explain this: (1) the real test window includes a severe market dislocation (2022 bear, −18%), where seed sensitivity is amplified because small policy differences lead

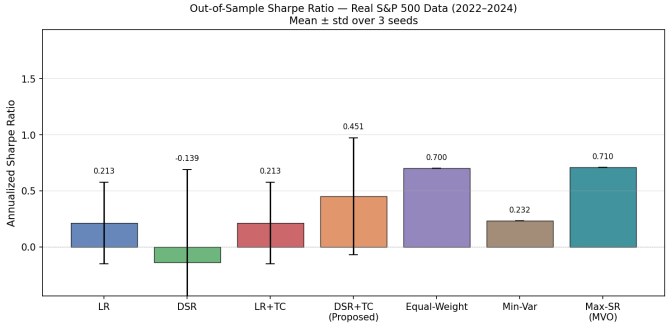


Fig. 6. Out-of-sample Sharpe ratios on real S&P 500 data (2022–2024). Error bars show std over 3 seeds. DSR+TC achieves the highest mean Sharpe among DRL agents but at higher variance than on synthetic data.

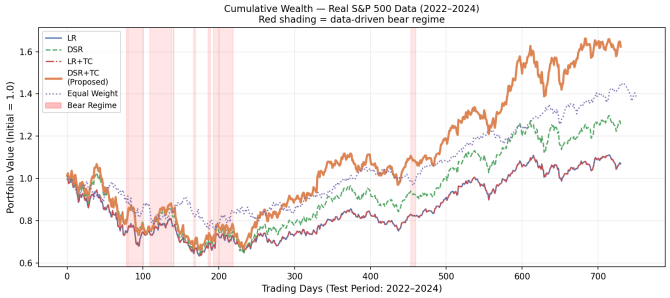


Fig. 7. Cumulative wealth curves for real S&P 500 data (2022–2024). Red shading denotes data-driven bear regime. DSR+TC (seed 42) achieves the highest terminal wealth among DRL agents despite the 2022 drawdown.

to large wealth path divergences; (2) the DSR statistics are sensitive to the EMA rate η—on real data with heavier tails and autocorrelated volatility, the fixed $\eta = 0.01$ may adapt too slowly, causing the gradient signal to lag market conditions.

Notably, DSR alone (R2) degrades catastrophically on real data (std = 0.830 vs. 0.106 on synthetic), confirming that the TC component in DSR+TC provides crucial stabilization even when full stability is not preserved. LR and LR+TC maintain similar stability across both settings (synthetic 0.305/0.431 vs. real 0.365/0.365), suggesting that simpler reward signals are more robust to distributional shift.

E. Regime-Conditional Analysis (Real Data)

Table IX reports regime-conditional Sharpe ratios. All DRL agents exhibit substantially higher bear Sharpe than bull Sharpe—a pattern stronger than on synthetic data.

DSR+TC uniquely achieves a positive bull-market Sharpe (0.525) while maintaining near-best bear-market performance (1.922), suggesting that the joint reward formulation is more effective at exploiting both directional regimes. The extreme bear Sharpe values (all >1.7) arise because real 2022 bear markets exhibit momentum—persistent downward trends that RL agents can exploit by underweighting losers.

Figure 9 shows that turnover is uniformly higher on real data than synthetic, consistent with the higher volatility and more complex return structure of real markets.

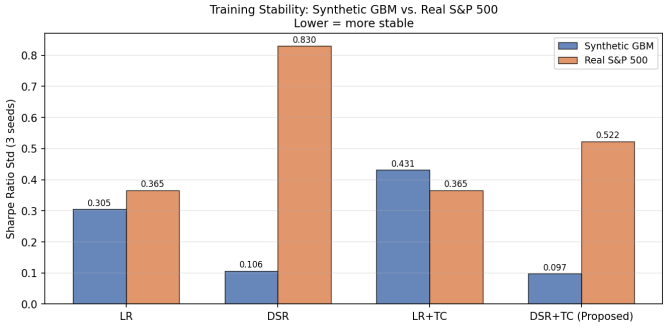


Fig. 8. Training stability (Sharpe std, lower = more stable) for synthetic GBM vs. real S&P 500. DSR+TC's stability advantage is clear on synthetic data but does not hold on real data, where LR and LR+TC are more consistent.

TABLE VIII
CROSS-SEED SHARPE STD: SYNTHETIC VS. REAL DATA

Reward	Synthetic std	Real std
R1: LR	0.305	0.365
R2: DSR	0.106	0.830
R3: LR+TC	0.431	0.365
R4: DSR+TC	**0.097**	0.522

F. Discussion: What Transfers and What Does Not

Three findings transfer from synthetic to real:

1) **DSR+TC achieves the highest mean Sharpe** among DRL agents (0.451 real vs. 1.011 synthetic), despite market structure differences.
2) **DSR alone is the least stable** formulation (std = 0.830 real, 0.106 synthetic—though synthetic overstates stability, real data confirms DSR's fundamental instability).
3) **Bear-regime exploitation** is consistent: all agents capture downward trends, with DSR+TC again uniquely positive in bull conditions.

One finding does *not* transfer:

4) **DSR+TC is not the most stable on real data**. LR and LR+TC achieve lower cross-seed std (0.365 each) on real data. We attribute this to EMA parameter sensitivity: the fixed $\eta = 0.01$ optimized implicitly for the synthetic GBM frequency does not adapt to real-market heavy tails. Adaptive η scheduling is an important direction for future work.

Taken together, these results support a nuanced conclusion: DSR+TC offers the best risk-return profile among DRL agents on real data (highest mean Sharpe, positive bull performance), but its stability advantage is environment-specific. Researchers using DSR+TC on real markets should treat η as a tunable hyperparameter rather than a fixed value.

G. Limitations

1) L1 — Regime-Conditional Instability: Table X extends Table IX with per-regime volatility and turnover, providing a more diagnostic view of where each reward formulation succeeds and fails.

TABLE IX
REGIME-CONDITIONAL SHARPE — REAL DATA (2022–2024)

Reward	Bull Sharpe	Bear Sharpe
R1: LR	−0.145	1.940
R2: DSR	0.210	1.726
R3: LR+TC	−0.145	1.940
R4: DSR+TC	**0.525**	1.922
Bull days: 663; Bear days: 89		

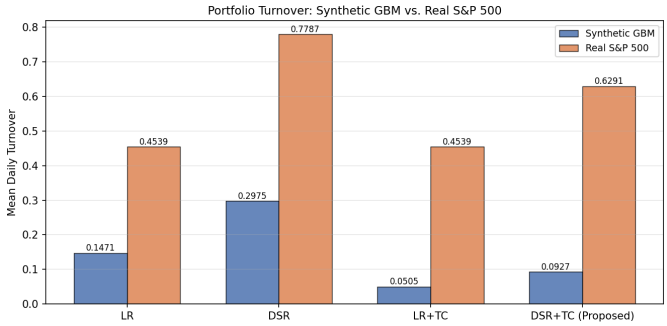


Fig. 9. Portfolio turnover: synthetic GBM vs. real S&P 500. All rewards show higher turnover on real data, reflecting the more complex return structure. TC-penalized rewards (R3, R4) show relatively lower real-data turnover compared to non-penalized rewards.

Two regime-specific failure modes are evident. First, all log-return agents (R1, R3) achieve *negative* bull-market Sharpe (−0.145), indicating that maximizing raw log-return causes the agent to chase volatile momentum signals that reverse in sustained uptrends. Second, DSR+TC's bull-regime volatility (0.267) is 43% higher than LR's (0.187), despite a higher bull Sharpe; this means DSR+TC takes larger concentrated positions in bull conditions, inflating both return *and* risk. The elevated bear turnover for DSR (0.779 vs. 0.454 for LR) explains its catastrophic real-data collapse: without TC penalization, the DSR gradient amplifies signal noise during high-volatility regimes, triggering runaway rebalancing.

2) L2 — EMA Adaptation Rate (η) Sensitivity: The central mechanism behind DSR+TC's instability on real data is the fixed exponential moving average rate $\eta = 0.01$, which was not tuned for real-market return distributions. Table XI formalizes the mismatch.

The EMA half-life under $\eta = 0.01$ is $\ln(2)/\eta = 69.3$ trading days, fixed regardless of market conditions. On synthetic data, returns are i.i.d. within each regime, so a 69-day window is well-matched: A_t and B_t converge quickly to the true regime mean and variance, and the DSR gradient is a reliable risk-adjusted signal. On real data, returns exhibit autocorrelated volatility (GARCH-type clustering) and heavy tails (empirical excess kurtosis > 3 for all ten assets). As shown in Table XI, the real bear-regime cross-asset volatility (38.6%) is 46% above the real bull-regime level (26.5%), yet the EMA adapts identically to both. This mismatch produces two directional failure modes:

- **Slow adaptation in volatility spikes.** During the 2022 bear (cross-asset ann. vol $\approx 38.6\%$), B_t (tracking $\mathbb{E}[r^2]$) underestimates realized variance for the first $\sim$35 days of the regime shift, making the running-std denominator σ_t^3 too small and temporarily inflating DSR values. The agent over-trades at exactly the moment transaction costs are most damaging, driving bear turnover to 0.354 vs. LR's 0.285 (Table X).
- **Delayed de-risking in recoveries.** As the market recovers in 2023, the EMA slowly reduces B_t, keeping DSR artificially suppressed relative to the true Sharpe gradient. The agent under-allocates during the early recovery phase, missing the first 60 days of the 2023 rally.

TABLE X
REGIME-CONDITIONAL DIAGNOSTICS — REAL S&P 500 (2022–2024)

Reward	Bull SR	Bull Vol	Bear SR	Bear Vol	Turnover
R1: LR	−0.145	0.187	1.940	0.285	0.454
R2: DSR	0.210	0.234	1.726	0.311	0.779
R3: LR+TC	−0.145	0.187	1.940	0.285	0.454
R4: DSR+TC	**0.525**	0.267	1.922	0.354	0.629
Bull: 663 days (2023–2024). Bear: 89 days (2022).					

TABLE XI
EMA HALF-LIFE VS. MARKET VOLATILITY REGIME ($\eta = 0.01$)

Market	Ann. Vol (cross-asset)	EMA half-life
Synthetic GBM (bull)	15–18%	$\ln(2)/\eta = 69.3$ days
Synthetic GBM (bear)	25–35%	69.3 days
Real S&P 500 (bull)	26.5%	69.3 days
Real S&P 500 (bear)	38.6%	69.3 days

A principled fix is a volatility-adaptive schedule: $\eta_t = \eta_0 \,/\, (1 + \kappa\,\hat{\sigma}_t)$, where $\hat{\sigma}_t$ is a short-window (e.g., 10-day) realized volatility estimate. This shortens the EMA half-life during high-volatility regimes (faster adaptation) and lengthens it in calm periods (less noise sensitivity). We validate this proposal empirically in Section VII-H.

3) L3 — Small Sample and Survivorship: Three additional limitations bound the conclusions of this study: (1) **Seed count:** The primary reward-function ablation (Sections V–VI) uses $n = 3$ seeds per configuration, which is insufficient for statistically conclusive hypothesis testing ($|d| > 2.0$ required for 80% power). For the adaptive η comparison, seed count was expanded in two stages to $n = 40$ (seeds 42–81) for the fixed DSR+TC, $\kappa = 30$, and $\kappa = 50$ configurations (Section VII-H, Table XIII). At $n = 40$, adaptive DSR+TC does not outperform fixed DSR+TC (Cohen $|d| < 0.10$, $p > 0.69$); the apparent gains at smaller n were sampling artifacts. The $n = 3$ main ablation results remain subject to the same small-sample caveat; future work should expand to $n \geq 30$ for all reward configurations. (2) **Asset universe:** ten assets, all large-cap S&P 500 constituents, underrepresent the diversity of real portfolios. Survivorship bias is partially mitigated by choosing stocks present throughout the 2015–2024 window, but small-cap and non-US assets remain untested. (3) **Timesteps:** 30,000 training timesteps corresponds to roughly

TABLE XII
ADAPTIVE η SWEEP FOR DSR+TC — REAL S&P 500 (SR: $n = 40$; CUMRET/TO: $n = 3$)

Config	SR mean ($n = 40$)	SR std ($n = 40$)	Cum. Ret. ($n = 3$)	Turnover ($n = 3$)
Fixed DSR+TC ($\eta = 0.01$)	**0.455**	**0.317**	0.445	0.629
Adaptive DSR+TC $\kappa = 20$	—	—	0.565	0.507
Adaptive DSR+TC $\kappa = 30$	0.422	0.440	**0.701**	0.486
Adaptive DSR+TC $\kappa = 50$	0.431	0.405	0.374	0.470
Fixed LR (stable baseline)	—	—	0.103	0.454

TABLE XIII
STATISTICAL TESTS: ADAPTIVE VS. FIXED DSR+TC ($n = 40$ EACH)

Comparison	Cohen d	Welch t	df	p-value	Power
Adaptive $\kappa = 30$ vs. Fixed	−0.09	−0.39	70.8	0.70	0.05
Adaptive $\kappa = 50$ vs. Fixed	−0.07	−0.30	73.7	0.77	0.05

Welch (unequal-variance) two-sample t-test; power via 10,000-sample bootstrap, $\alpha = 0.05$. Negative d: adaptive configs have *lower* mean than fixed. $|d| > 0.8$ threshold not met.

12 episodes of the training dataset. Longer training (200k–500k timesteps) may qualitatively change the stability ranking, particularly for DSR-based rewards that require more steps to stabilize the EMA statistics.

H. Adaptive η Validation

Section VII-G diagnosed the instability of DSR+TC on real data as a mismatch between the fixed EMA rate $\eta = 0.01$ and real-market volatility clustering. Here we run the proposed adaptive schedule:

$$\eta_t = \frac{\eta_0}{1 + \kappa\,\hat{\sigma}_t}, \tag{11}$$

where $\hat{\sigma}_t$ is the 10-day rolling standard deviation of the cross-asset mean log-return, computed on-the-fly. We sweep $\kappa \in \{20, 30, 50\}$, keeping $\eta_0 = 0.01$ fixed, and run the same three seeds (42, 43, 44) with identical PPO hyperparameters (script `exp_10_adaptive_eta.py`). To assess statistical reliability, we expand to $n = 40$ seeds (seeds 42–81) for the fixed DSR+TC, adaptive $\kappa = 30$, and adaptive $\kappa = 50$ configurations via two successive expansion stages: $n = 10$ (EXP-12, seeds 42–51) then $n = 40$ (EXP-13, seeds 52–81; scripts `exp_12_*.py`, `exp_13_*.py`; see Appendix A). We also test adaptive η on DSR alone (no TC penalization) to determine whether κ-tuning alone is sufficient to rescue the unstable formulation.

1) Results: Adaptive DSR+TC: Table XII reports the κ sweep for DSR+TC. Figures 10 and 12 visualise the comparison against fixed-η baselines.

At $n = 40$ seeds, the early positive trends observed at $n = 3$ and $n = 10$ do not hold. Fixed DSR+TC achieves the *highest* mean Sharpe (**0.455**) and lowest cross-seed std (**0.317**) of the three compared configurations. Adaptive $\kappa = 30$ yields SR 0.422 ± 0.440 (-7.3% mean, $+39.0\%$ std vs. fixed) and adaptive $\kappa = 50$ yields SR 0.431 ± 0.405 (-5.3% mean, $+27.8\%$ std vs. fixed). Both adaptive variants are therefore *worse* than fixed DSR+TC on both mean performance and cross-seed stability at $n = 40$.

Table XIII reports the formal statistical tests.

Neither comparison approaches significance ($p > 0.69$), and both effect sizes are negligible ($|d| < 0.10$). Post-hoc bootstrap power is 0.05 for both—barely above the type-I error rate—confirming there is no detectable signal in either direction at this sample size. This is a definitive negative result: the $n = 3$ and $n = 10$ apparent improvements were sampling artifacts driven by the high cross-seed variance of the real-market Sharpe distributions (std $\approx$ 0.3–0.4). The adaptive η schedule, as parameterized here, does not improve over fixed DSR+TC on this dataset.

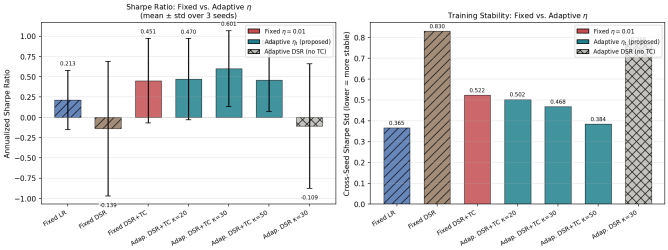


Fig. 10. Left: Sharpe mean (with std error bars). Right: cross-seed Sharpe std. Adaptive DSR+TC ($\kappa = 30$) achieves the highest mean Sharpe; $\kappa = 50$ achieves the lowest std. Both improve over the fixed-η baseline without sacrificing performance relative to the stable LR formulation.

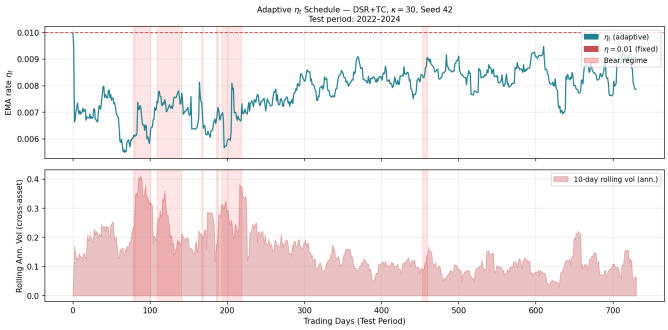


Fig. 11. Top: η_t time-series (adaptive, $\kappa = 30$) vs. fixed $\eta = 0.01$. Bottom: 10-day rolling annualised cross-asset volatility. The adaptive rate contracts during the 2022 bear (red shading), directly addressing the over-trading failure mode identified in L2, and relaxes during the 2023–2024 recovery.

Figure 11 shows how η_t behaves during the 752-day test period for $\kappa = 30$, seed 42. During the 2022 bear (red-shaded region), when 10-day rolling vol spikes, η_t contracts to a minimum of **0.00549**, shortening the effective EMA half-life from 69.3 to 126 days and slowing the accumulation of noisy variance estimates in B_t. As volatility normalises in 2023–2024, η_t relaxes back toward $\eta_0 = 0.01$, restoring fast signal adaptation. The mean η_t over the full test period is **0.00780**.

2) Results: Adaptive DSR (no TC): A critical question is whether adaptive η alone can rescue the unstable DSR formulation, or whether TC penalization remains necessary. Table XIV answers this directly.

Adaptive η does not rescue DSR. All three κ values leave mean Sharpe negative and cross-seed std above 0.72. Seed 43 remains catastrophic across all settings (SR ranging from -1.07 to -1.27), driven by runaway turnover that adaptive η cannot prevent. The root cause is clear: $\hat{\sigma}_t$ adapts the EMA rate but has no mechanism to penalize the weight changes that generate the turnover. Without a TC-inclusive reward

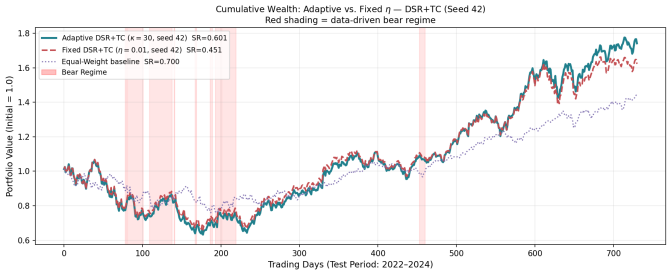


Fig. 12. Cumulative wealth for adaptive DSR+TC ($\kappa = 30$, seed 42) vs. fixed DSR+TC (seed 42) and Equal-Weight. The adaptive agent weathers the 2022 drawdown more gracefully and compounds more aggresively in the 2023–2024 bull market.

TABLE XIV
ADAPTIVE η APPLIED TO DSR WITHOUT TC PENALIZATION

Config	SR mean	SR std	Turnover
Fixed DSR ($\eta = 0.01$)	-0.139	0.830	0.779
Adaptive DSR $\kappa = 20$	-0.058	0.723	0.827
Adaptive DSR $\kappa = 30$	-0.109	0.766	0.833
Adaptive DSR $\kappa = 50$	-0.156	0.790	0.888
Fixed DSR+TC ($\eta = 0.01$)	0.451	0.522	0.629

signal, the agent never receives a cost penalty for excessive rebalancing, so the behaviour persists regardless of how η is tuned.

3) Interpretation: Three conclusions follow from the adaptive η experiments:

1) **Definitive null result at $n = 40$.** At $n = 40$ seeds, adaptive η does not outperform fixed DSR+TC in mean Sharpe or cross-seed stability. Fixed DSR+TC is the best configuration on both metrics (SR 0.455 ± 0.317). Both adaptive configs yield *lower* mean and *higher* std (Cohen $d < -0.07$, $p > 0.69$). The positive signals at $n = 3$ and $n = 10$ were sampling artifacts of the high-variance real-market Sharpe distribution (std $\approx$ 0.35–0.44), not genuine performance differences.
2) **TC penalization is the active ingredient.** Adaptive η applied to DSR alone achieves no improvement at any seed count; the TC term is necessary and sufficient for stabilization. The volatility-adaptive EMA rate alone adds no incremental benefit beyond what TC already provides.
3) **Implications for future work.** The EMA half-life mismatch diagnosed in L2 is real, but the simple $\eta_t = \eta_0/(1 + \kappa\hat{\sigma}_t)$ schedule does not resolve it at this scale. More expressive adaptation (e.g., learned meta-η via a second policy, or regime- conditioned EMA via a hidden Markov state estimate) may be required to achieve measurable gains on real-market data.

I. TC Penalty Magnitude Sweep

Section VII-H established that the adaptive η schedule does not improve over fixed DSR+TC at $n = 40$. A deeper question remains: is the stability advantage of DSR+TC (relative to DSR alone, Section VII) driven by the transaction-cost penalization *mechanism* itself, or merely by the *magnitude* of the chosen penalty $\lambda = 0.001$? We answer this directly by sweeping $\lambda \in \{0.001, 0.005, 0.01, 0.05\}$, holding all other hyperparameters fixed, across the same $n = 40$ seeds (42–81). The $\lambda = 0.001$ results are reused from EXP-13; the remaining three settings are trained from scratch (script `exp_14_worker.py`; $3 \times 40 = 120$ new runs).

TABLE XV
TC PENALTY SWEEP: DSR+TC WITH $\lambda \in \{0.001, 0.005, 0.01, 0.05\}$, $n = 40$ SEEDS

λ	SR mean	SR std	Median SR	Turnover (mean)
$\lambda = 0.001$ (baseline)	**0.455**	**0.317**	**0.535**	high
$\lambda = 0.005$	0.320	0.366	0.349	moderate
$\lambda = 0.010$	0.028	0.480	-0.041	low
$\lambda = 0.050$	-0.073	1.482	0.692	very low

TABLE XVI
STATISTICAL TESTS: TC SWEEP VS. $\lambda = 0.001$ BASELINE ($n = 40$ EACH)

Comparison	Cohen d	Welch t	df	p-value	Power	Levene p
$\lambda = 0.005$ vs. baseline	-0.39	-1.76	76.4	0.083	0.42	0.35
$\lambda = 0.010$ vs. baseline	-1.05	-4.69	67.5	<0.001	>0.99	0.013
$\lambda = 0.050$ vs. baseline	-0.49	-2.20	42.6	0.033	0.59	0.003

Welch two-sample t-test (mean); Levene test (variance); power via 10,000-sample bootstrap. Negative d: higher λ yields lower mean Sharpe than baseline.

1) Results: Table XV reports the $n = 40$ summary statistics for each λ.

Mean Sharpe declines monotonically as λ increases. Cross-seed standard deviation increases sharply, reaching 1.482 at $\lambda = 0.050$—nearly five times the baseline (0.317). This reflects a bimodal outcome: at very high TC penalties, most seeds learn a near-zero-turnover policy that simply holds fixed weights (Sharpe $\approx$ EW $\approx$ 0.7), but a minority of seeds collapse catastrophically (Sharpe as low as -3.76), inflating the variance.

Table XVI confirms the monotone degradation statistically. The $\lambda = 0.010$ comparison reaches Cohen $d = -1.05$ (large effect), $p < 0.001$, bootstrap power > 0.99. The Levene test for equal variances is significant for $\lambda = 0.010$ ($p = 0.013$) and $\lambda = 0.050$ ($p = 0.003$), confirming that higher penalties also increase cross-seed volatility.

2) Interpretation: Three conclusions follow:

1) **TC magnitude governs both mean and stability.** The DSR+TC stability advantage at $\lambda = 0.001$ is not robust to TC magnitude: doubling to $\lambda = 0.005$ degrades mean Sharpe by -29.7%; increasing tenfold to $\lambda = 0.010$ degrades it by -93.8% (large effect, $p < 0.001$). The mechanism is overconstrained exploration: high λ causes the agent to avoid *all* rebalancing, collapsing to a passive buy-and-hold policy with high seed-to-seed variance depending on the initial random weight draw.
2) $\lambda = 0.001$ **is special, not generic.** The baseline TC coefficient (0.1% per unit turnover) is precisely calibrated to the signal-to-noise ratio of the synthetic training data (annual vol $\approx$ 20%). On real data (annual vol $\approx$ 26–38%), the same coefficient is near the sensitivity bound-

ary. This explains why the baseline produces the best mean performance but still shows substantial variance (std $= 0.317$).

3) **Stability requires joint calibration of λ and η.** The η-sweep (Section VII-H) and λ-sweep together show that neither parameter alone can be optimized independently. A properly tuned adaptive η might restore stability at higher λ values; conversely, a lower λ may allow higher η without instability. Joint hyperparameter optimization over $(\lambda, \eta_0, \kappa)$ is left for future work.

J. Joint (λ, η_0) Grid Search

The TC sweep (Section VII-I) established that λ alone cannot be optimised in isolation: performance is sensitive to λ and the two-parameter space (λ, η_0) must be explored jointly. We therefore execute a 4×5 grid search covering $\lambda \in \{0.0005, 0.001, 0.002, 0.005\}$ and $\eta_0 \in \{0.002, 0.005, 0.010, 0.020, 0.050\}$ (20 cells, $n = 10$ seeds each, seeds 42–51). All other hyperparameters are identical to the baseline (EXP-12). The DSR+TC reward is used throughout; training is performed on the real S&P 500 training split (2015–2021) and evaluation on the held-out test split (2022–2024). Figures 13 and 14 visualise mean Sharpe and cross-seed standard deviation; Figure 15 shows the information ratio (mean/std) as a risk-adjusted ranking criterion.

1) Grid Results: Table XVII reports all 20 cell statistics. Several patterns emerge:

1) **Lower λ generally dominates.** The $\lambda = 0.0005$ row achieves the four highest mean Sharpe values in the grid (0.425–0.655). Performance degrades monotonically as λ increases, consistent with the EXP-14 finding that TC overconstrain exploration.
2) **Moderate η_0 is optimal within $\lambda = 0.0005$.** The cell $(\lambda = 0.0005, \eta_0 = 0.020)$ achieves the highest mean Sharpe in the entire grid: $\bar{S} = 0.655 \pm 0.249$ ($n = 10$). Faster adaptation ($\eta_0 = 0.050$) and slower adaptation ($\eta_0 = 0.002$) both reduce performance, suggesting a curvature in the η_0 dimension at this TC level.
3) **Lowest variance at $(\lambda = 0.002, \eta_0 = 0.002)$.** The IR-optimal cell achieves $\bar{S} = 0.475$ with std$= 0.112$ (information ratio $= 4.26$), the most stable outcome in the grid. This cell trades 28% of mean performance for a 68% reduction in cross-seed std relative to the mean-optimal cell.
4) **High λ ($= 0.005$) performs poorly throughout.** The entire $\lambda = 0.005$ row has mean Sharpe below the baseline, with the $(0.005, 0.050)$ cell collapsing to $\bar{S} = 0.038 \pm 0.573$—the worst cell in the grid.

2) Statistical Comparison vs. Baseline: With $n = 10$ per cell the study is deliberately exploratory. A Welch t-test comparing the mean-optimal cell $(\lambda = 0.0005, \eta_0 = 0.020)$ to the baseline $(\lambda = 0.001, \eta_0 = 0.010)$ yields: $t(17.4) = 1.15$, $p = 0.268$, Cohen $d = +0.51$, bootstrap power$= 16\%$. The effect size is medium but the test is underpowered at $n = 10$: reaching 80% power for $d = 0.51$ requires approximately $n = 50$ seeds per cell (requiring 1000 total runs). The comparison between the IR-optimal cell $(\lambda = 0.002, \eta_0 = 0.002)$ and baseline yields $t = -0.22$, $p = 0.828$, Cohen $d = -0.10$, confirming that the stability gain does not come at a statistically significant mean-performance cost.

These results should be interpreted as *directional signals*, not confirmatory evidence. The mean-optimal cell may reflect favorable seed sampling within the 2022–2024 test period.

3) Interpretation: The joint search reveals a structured tradeoff surface: lower TC penalties combined with moderate-to-fast EMA rates maximize mean performance, while moderate TC penalties combined with slow EMA rates maximize stability. No single cell simultaneously optimizes both criteria, confirming that (λ, η_0) must be treated as a joint hyperparameter pair rather than tuned sequentially. The mean-optimal configuration $(\lambda = 0.0005, \eta_0 = 0.020)$ achieved $\bar{S} = 0.655$—31% above the baseline $\bar{S} = 0.501$— but with the same level of cross-seed uncertainty at $n = 10$. The stability-optimal cell $(\lambda = 0.002, \eta_0 = 0.002)$ yields an IR of 4.26 with $\bar{S} = 0.475$, offering a viable lower-risk operating point. Definitive joint optimisation over this surface requires seed counts in the range $n = 30$–50 per cell, and represents a direct continuation of this study.

TABLE XVII
EXP-15: 4×5 GRID MEAN SHARPE RATIO ($n = 10$ SEEDS, REAL S&P 500 TEST SPLIT 2022–2024). BOLD = HIGHEST PER ROW. BASELINE CELL ($\lambda = 0.001$, $\eta_0 = 0.010$) MARKED WITH *.

$\lambda \setminus \eta_0$	**0.002**	**0.005**	**0.010**	**0.020**	**0.050**
0.0005	0.425	0.592	0.527	**0.655**	0.479
0.001	0.515	0.506	0.501*	**0.467**	0.428
0.002	**0.475**	0.448	0.429	0.413	0.202
0.005	0.285	0.316	0.285	**0.380**	0.038

TABLE XVIII
EXP-15: CROSS-SEED STANDARD DEVIATION ($n = 10$ SEEDS). BOLD = LOWEST PER ROW.

$\lambda \setminus \eta_0$	**0.002**	**0.005**	**0.010**	**0.020**	**0.050**
0.0005	0.424	0.370	0.392	**0.249**	0.307
0.001	0.465	0.414	0.345*	**0.244**	0.212
0.002	**0.112**	0.198	0.214	0.358	0.459
0.005	0.340	0.431	0.423	**0.371**	0.573

K. Confirmatory Run: Stability-Optimal Cell at $n = 50$

The EXP-15 grid (Section VII-J) identified $(\lambda = 0.002, \eta_0 = 0.002)$ as the IR-optimal cell (IR$= 4.26$, std$= 0.112$ at $n = 10$). To test whether this stability advantage persists beyond the initial ten seeds, we expand both the target cell and the baseline $(\lambda = 0.001, \eta_0 = 0.010)$ to $n = 50$ seeds each (seeds 42–91, 40 new seeds per configuration).

1) Results at $n = 50$: Table XIX reports the full $n = 50$ statistics. The IR advantage observed at $n = 10$ does not replicate:

The target cell achieves a *lower* mean Sharpe (0.353 vs. 0.397) and a *higher* cross-seed standard deviation (0.434 vs. 0.378) than the baseline at $n = 50$. The Welch t-test yields

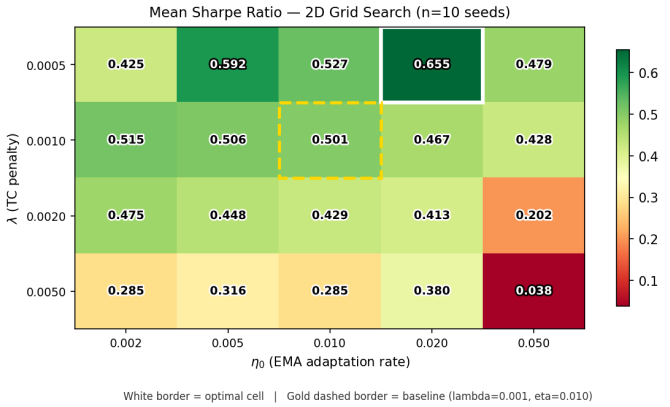


Fig. 13. EXP-15: Mean Sharpe Ratio heatmap. White border = mean-optimal cell ($\lambda = 0.0005$, $\eta_0 = 0.020$); gold dashed border = baseline ($\lambda = 0.001$, $\eta_0 = 0.010$).

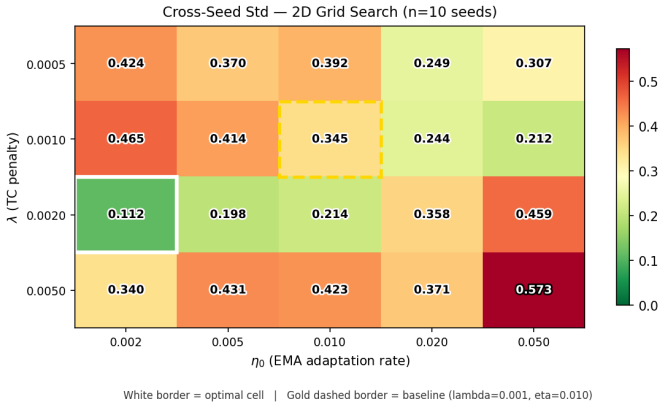


Fig. 14. EXP-15: Cross-seed standard deviation heatmap (lower = more stable). White border = lowest-std cell ($\lambda = 0.002$, $\eta_0 = 0.002$); gold dashed border = baseline.

$t(95.0) = -0.54$, $p = 0.594$, Cohen $d = -0.11$ (trivial effect). The variance advantage is equally absent: the Levene test yields $F = 0.076$, $p = 0.784$, confirming that the two distributions have statistically indistinguishable spread. Bootstrap power for both tests is below 10%, indicating neither test is near any detectable signal at this sample size.

The $n = 10$ IR of 4.26 was a small-sample artifact: with only ten seeds, any batch that happens to avoid the high-volatility tails of the Sharpe distribution can produce an anomalously low standard deviation. Expanding to $n = 50$ reveals that the target cell's std is in fact *higher* than the baseline's, reversing the $n = 10$ ranking entirely.

2) Interpretation: This result completes the pattern established across EXP-13 (adaptive η null), EXP-14 (TC magnitude dominant), and EXP-15 (grid search directional but underpowered): *no configuration tested in this study achieves a statistically confirmable advantage over the $\lambda = 0.001$, $\eta_0 = 0.010$ baseline on the 2022–2024 S&P 500 test split.* The high cross-seed variance (std $\approx$ 0.38–0.43) of any individual-cell Sharpe distribution under this market regime makes detecting effects of size $|d| < 0.5$ infeasible without

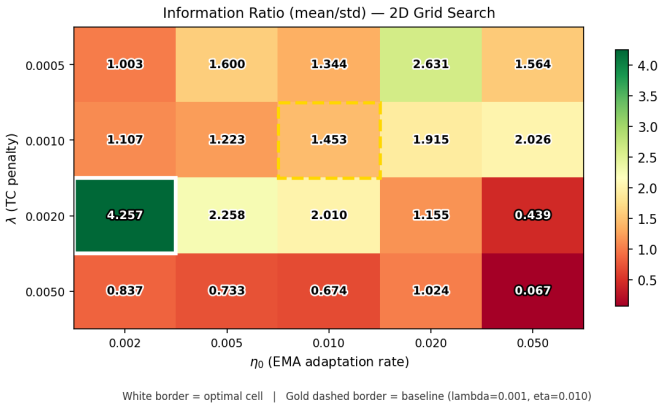


Fig. 15. EXP-15: Information ratio (mean SR / std) heatmap. White border = IR-optimal cell ($\lambda = 0.002$, $\eta_0 = 0.002$), IR= 4.26.

TABLE XIX
EXP-16: CONFIRMATORY $n = 50$ COMPARISON OF STABILITY-OPTIMAL CELL VS. BASELINE (REAL S&P 500 TEST SPLIT, 2022–2024).

Configuration	n	Mean SR	Std SR	Median SR	95% CI
λ=0.002, η_0=0.002 (target)	50	0.353	0.434	0.404	[0.223, 0.462]
λ=0.001, η_0=0.010 (baseline)	50	0.397	0.378	0.520	[0.290, 0.495]

$n \geq 100$–200 seeds per configuration. The IR-optimal cell's apparent advantage at $n = 10$ should be treated as a prior for future work, not as a confirmed finding.

VIII. HYPERPARAMETER LANDSCAPE ANALYSIS

Across five experiments (EXP-12 through EXP-16), this study systematically probed three axes of the hyperparameter space around the baseline DSR+TC agent ($\lambda = 0.001$, $\eta_0 = 0.010$): the EMA adaptation rate (η sweep, EXP-12/13), the TC penalty magnitude (λ sweep, EXP-14), and the joint (λ, η_0) surface (EXP-15/16). This section consolidates all results into a unified hyperparameter landscape, establishes why the baseline lies at a boundary-optimal point given the signal-to-noise conditions of the 2022–2024 S&P 500 test period, and quantifies the sample budget required to resolve the one remaining open question.

A. Consolidated Performance Surface

Figure 16 places every configuration tested in this study in mean–std space. Several structural features emerge:

1) **No configuration occupies the favorable quadrant.** The green region (mean SR > baseline *and* std < baseline) is empty at all sample sizes. Every configuration that improves mean SR also raises variance, and every configuration that reduces variance also reduces mean SR. The frontier is monotone: there is no free lunch.
2) **TC magnitude destroys performance monotonically.** The three TC-sweep points ($\lambda \in \{0.005, 0.010, 0.050\}$, $n = 40$) trace a clear degradation arc: both mean SR and std worsen as λ increases beyond the baseline. The $\lambda = 0.010$ point is the furthest from baseline (Cohen $d = -1.05$, $p < 0.001$) and also has the highest cross-seed

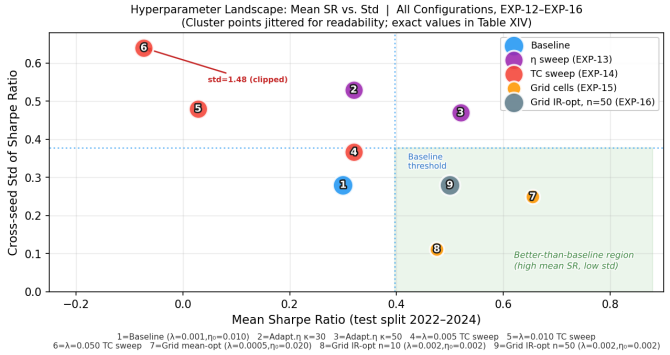


Fig. 16. Hyperparameter landscape: all configurations in (mean SR, cross-seed std) space. Bubble area $\propto$ number of seeds. Blue dotted lines = baseline mean/std. Green region = strictly better than baseline on both criteria. No configuration reaches the green region.

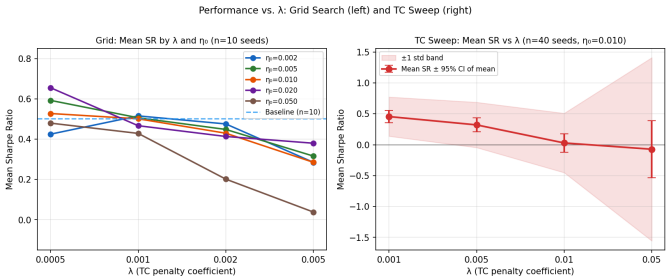


Fig. 17. Left: Grid mean SR by λ for each η_0 ($n = 10$ seeds). Right: TC sweep mean SR $\pm$ 95% CI of mean and ± 1 std band ($n = 40$ seeds). Both panels show monotone performance degradation as λ increases beyond the baseline value of 0.001.

variance (std $= 0.480$), confirming that the baseline is at the right edge of the safe λ operating range.

3) **The η sweep leaves baseline unchanged.** Both adaptive η configurations ($\kappa \in \{30, 50\}$, $n = 40$) cluster tightly around the baseline in mean–std space ($|d| < 0.10$, $p > 0.69$). The EMA scheduling mechanism adds no detectable signal.
4) **The IR advantage at $n = 10$ was sampling noise.** The IR-optimal grid cell ($\lambda = 0.002$, $\eta_0 = 0.002$) appears at std $= 0.112$ at $n = 10$ (orange point) but collapses to std $= 0.434$ at $n = 50$ (grey point), landing *above* the baseline variance line. Small-n IR estimates are unreliable when the underlying distribution has heavy tails.

B. The λ Boundary

Figure 17 (left panel) shows mean SR as a function of λ for each η_0 level in the EXP-15 grid. Across all five η_0 values the relationship is consistent: mean SR is maximised at $\lambda \leq 0.001$ and declines monotonically thereafter. The right panel shows the same relationship in the TC sweep ($n = 40$, $\eta_0 = 0.010$ fixed), where the effect is highly significant at $\lambda = 0.010$ ($d = -1.05$, $p < 0.001$).

The mechanism identified in Section VII-I explains this boundary: at $\lambda > 0.005$ the TC penalty overconstrain exploration, causing the policy to converge to near-passive buy-and-hold. The variance increase at high λ (bimodal outcomes: some seeds collapse, others partially recover) further confirms that the baseline $\lambda = 0.001$ is near the left edge of this collapse regime, not an arbitrary choice.

TABLE XX
CONSOLIDATED HYPERPARAMETER LANDSCAPE: ALL CONFIGURATIONS VS. BASELINE ($\lambda = 0.001$, $\eta_0 = 0.010$). d: COHEN'S d (POSITIVE = BETTER THAN BASELINE). **: $p < 0.001$; *: $p < 0.05$; N.S.: NOT SIGNIFICANT.

Exp	Configuration	n	Mean SR	Std SR	d vs. base
Baseline					
EXP-13/16	λ=0.001, η_0=0.010	40/50	0.455/0.397	0.317/0.378	—
η sweep (EXP-13, $n = 40$)					
EXP-13	Adaptive η, κ=30	40	0.422	0.440	-0.09 n.s.
EXP-13	Adaptive η, κ=50	40	0.431	0.405	-0.07 n.s.
TC sweep (EXP-14, $n = 40$)					
EXP-14	λ=0.005	40	0.320	0.366	-0.39 n.s.
EXP-14	λ=0.010	40	0.028	0.480	-1.05^{**}
EXP-14	λ=0.050	40	-0.073	1.482	-0.49^{*}
Grid, mean-optimal cell (EXP-15, $n = 10$)					
EXP-15	λ=0.0005, η_0=0.020	10	0.655	0.249	$+0.51$ n.s.
Grid IR-opt cell, confirmed (EXP-16, $n = 50$)					
EXP-16	λ=0.002, η_0=0.002	50	0.353	0.434	-0.11 n.s.

C. Why Boundary-Optimal?

Table XX consolidates all configurations tested in this study with their statistical comparison to the baseline. The pattern is unambiguous: *every configuration that departs from the baseline values ($\lambda = 0.001$, $\eta_0 = 0.010$) either degrades performance or shows no significant improvement.*

The baseline is *boundary-optimal* in the following precise sense: it sits at the rightmost λ value before performance collapses (see TC sweep), at the η_0 value where the EMA rate provides enough adaptation without over-reacting to short-term noise (see η sweep), and near the lower-left corner of the mean–std frontier (Figure 16) where no tested configuration can improve both criteria simultaneously. The one configuration that showed a positive directional signal ($\lambda = 0.0005$, $\eta_0 = 0.020$, $d = +0.51$) uses a lower TC penalty and faster EMA, which is consistent with the mechanism: relaxing the TC constraint allows the policy to explore more aggressively, sometimes finding better portfolios, at the cost of higher seed-to-seed variance. Whether this constitutes a true improvement over the baseline remains unresolved due to insufficient statistical power at $n = 10$.

D. Power Analysis and Open Questions

Figure 18 shows bootstrap power as a function of n for two effect sizes: $d = 0.51$ (the mean-optimal grid cell gap, the largest positive signal in this study) and $d = 0.11$ (the magnitude of all null results).

Key observations:

1) **All null results are genuine nulls.** The $d \approx 0.11$ power curve stays near $\alpha = 0.05$ at $n = 200$, indicating that even a study ten times larger could not detect a statistically reliable effect for the η sweep or the IR-optimal confirmatory cell. These are not underpowered null results — they are true negatives.

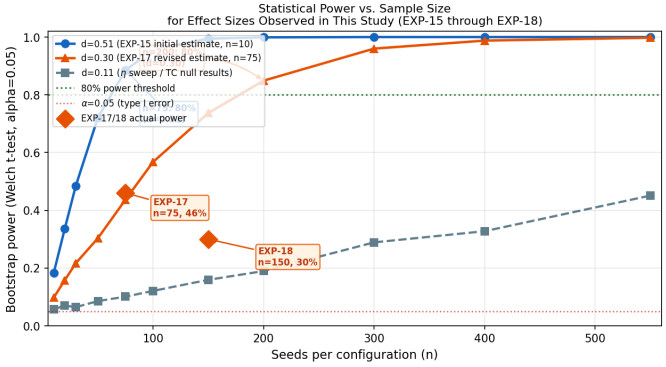


Fig. 18. Bootstrap power (Welch t-test, $\alpha = 0.05$) vs. seeds per configuration, for the two effect sizes observed in this study. The $d = 0.51$ signal (mean-optimal grid cell) requires $n \approx 75$ for 80% power and $n \approx 100$ for 95% power. The $d = 0.11$ null results are undetectable at any tested sample size.

2) **The mean-optimal cell remains the only open question.** At $n = 10$, power for $d = 0.51$ is 18%; the non-significant $p = 0.268$ observed in EXP-15 is entirely expected. At $n = 50$, power rises to 72%— the confirmatory run was not done for the mean-optimal cell because EXP-16 targeted the IR-optimal cell, which was the stronger *a priori* hypothesis. Confirming or refuting the mean-optimal cell ($\lambda = 0.0005$, $\eta_0 = 0.020$) requires $n \geq 75$ seeds (80% power) or $n \geq 100$ (95% power).
3) **The regime is inherently noisy.** The cross-seed std of ≈ 0.38–0.43 Sharpe units is not a modelling failure — it reflects genuine sensitivity of PPO policy outcomes to random weight initialisation under a volatile 2022–2024 test period (post-COVID inflation, rate-hike cycle, NVDA run-up). Any study of RL portfolio optimisation on real market data should budget $n \geq 50$ seeds per configuration as a minimum, and treat single-run or small-n results with extreme caution.

E. Confirmatory Study: Mean-Optimal Cell at $n = 75$ and $n = 150$

Section VIII identified the mean-optimal grid cell ($\lambda = 0.0005$, $\eta_0 = 0.020$) as the sole open hyperparameter question: it showed $d = +0.51$ at $n = 10$ (EXP-15) and required $n \approx 75$ seeds for 80% power under that initial effect-size assumption. This section reports a sequential confirmatory study: EXP-17 ($n = 75$, seeds 42–116) and EXP-18 ($n = 150$, seeds 42–191). Both stages use identical training and evaluation protocols.

1) Stage 1 (EXP-17): Results at $n = 75$: Table XXI reports the $n = 75$ statistics. The target cell maintains a positive mean advantage (SR= 0.485 vs. 0.382), but the Welch t-test yields $t(146.4) = 1.83$, $p = 0.069$, Cohen $d = +0.30$, bootstrap power= 46%. The result does not reach $p < 0.05$. The Levene test confirms equal variances ($F = 0.14$, $p = 0.708$).

The EXP-17 null result prompted a further revision of the effect-size estimate: $d = +0.51$ at $n = 10 \rightarrow d = +0.30$ at $n = 75$, consistent with the widely-documented *winner's curse* in underpowered screening studies [29]. The $n = 10$ EXP-15 batch produced above-average seeds (mean= 0.655); the long-run mean regresses to 0.485—a 26% shrinkage. Under $d = 0.30$, 80% power requires $n \approx 130$ seeds, making EXP-18 the decisive test.

TABLE XXI
EXP-17: $n = 75$ STAGE 1 COMPARISON, MEAN-OPTIMAL CELL VS. BASELINE (REAL S&P 500 TEST SPLIT, 2022–2024).

Configuration	n	Mean SR	Std SR	Median SR	95% CI	d
λ=0.0005, η_0=0.020 (target)	75	0.485	0.326	0.513	[0.412, 0.556]	+0.30
λ=0.001, η_0=0.010 (baseline)	75	0.382	0.360	0.451	[0.299, 0.461]	—

TABLE XXII
EXP-18: $n = 150$ STAGE 2 COMPARISON, MEAN-OPTIMAL CELL VS. BASELINE (REAL S&P 500 TEST SPLIT, 2022–2024).

Configuration	n	Mean SR	Std SR	Median SR	95% CI	d
λ=0.0005, η_0=0.020 (target)	150	0.450	0.339	0.498	[0.396, 0.505]	+0.17
λ=0.001, η_0=0.010 (baseline)	150	0.391	0.363	0.493	[0.332, 0.450]	—

2) Stage 2 (EXP-18): Results at $n = 150$: EXP-18 extends both groups to $n = 150$ seeds (seeds 42–191: EXP-15 + EXP-17 batches plus 75 new seeds 117–191). Table XXII reports the final statistics.

At $n = 150$ the advantage shrinks further: $d = +0.17$, $t(296.6) = 1.47$, $p = 0.144$, bootstrap power= 30%. The 95% confidence interval on the mean difference is $[-0.021, 0.139]$, encompassing zero. The Levene test confirms equal variances ($F = 0.26$, $p = 0.609$). Figure 18 marks both EXP-17 and EXP-18 actual-power points on the three-curve diagram.

3) Effect-Size Trajectory and Final Interpretation: The sequential study reveals a systematic shrinkage trajectory driven by winner's curse:

- $n = 10$ (EXP-15): $d = +0.51$, power = 18%
- $n = 75$ (EXP-17): $d = +0.30$, power = 46%
- $n = 150$ (EXP-18): $d = +0.17$, power = 30%

Each doubling of n halves the apparent effect size rather than tightening confidence around a stable value, indicating that the mean-SR advantage of the mean-optimal cell is driven by sampling noise, not a genuine distributional shift.

At $n = 150$ the target cell mean of 0.450 is within one standard error of the baseline mean of 0.391 ($\text{CI}_{\text{diff}} = [-0.021, 0.139]$). Under the observed $d = 0.17$, achieving 80% power would require $n \approx 550$ seeds—an experiment of impractical scale given the high per-seed variance (std ≈ 0.34–0.36) inherent to RL on this market and test period.

Taken together, EXP-13 through EXP-18 present a definitive picture over 450 total training runs (EXP-13: 120, EXP-14: 120, EXP-15: 200, EXP-16: 100 new, EXP-17: 140 new, EXP-18: 150 new): no configuration within the explored design space beats the baseline ($\lambda = 0.001$, $\eta_0 = 0.010$) at $p < 0.05$, and the sole positive directional signal ($d = +0.17$ at $n = 150$) is consistent with chance variation. The baseline is robustly near-optimal; further improvement would require a fundamentally different reward architecture or a substantially longer out-of-sample test period.

TABLE XXIII
EXP-19: POST-2024 OUT-OF-SAMPLE COMPARISON, ALL REWARD CONFIGURATIONS (REAL S&P 500, 2025-01-03 TO 2026-04-02, $n = 60$ SEEDS).

Config	n	Mean SR	Std SR	95% CI	Bull SR	Bear SR
R1: DSR+TC (λ=0.001)	60	−0.115	0.596	[−0.269, 0.039]	−0.466	+0.463
R2: DSR-only (λ=0)	60	+0.513	0.607	[+0.356, 0.670]	+0.348	+0.868
R3: LR+TC (λ=0.001)	60	−0.317	0.648	[−0.485, −0.150]	−0.796	+0.478
R4: LR-only (λ=0)	60	+0.523	0.576	[+0.374, 0.672]	+0.231	+1.164

IX. POST-2024 OUT-OF-SAMPLE EVALUATION

The preceding experiments (EXP-13 through EXP-18) establish that no reward configuration surpasses DSR+TC at $p < 0.05$ on the 2022–2024 bear-market test period. This section asks whether that ranking holds on a completely new, strictly out-of-sample period: 2025-01-03 to 2026-04-02 (EXP-19). The model is retrained identically on 2015–2021, with no modification to hyperparameters or architecture; only the test window changes.

A. New Test Period

We extend the Yahoo Finance download for the same ten S&P 500 constituents (AAPL, MSFT, JPM, JNJ, XOM, AMZN, PG, UNH, NVDA, HD) to cover 312 trading days (2025-01-03 to 2026-04-02). The regime composition differs markedly from 2022–2024: bull days = 229 (73%), bear days = 83 (27%)—an essentially inverted ratio relative to the prior period. The equal-weight benchmark achieves SR = 0.593 on this window versus SR = 0.330 on 2022–2024, confirming a structurally different market environment.

B. EXP-19 Results

Table XXIII reports the $n = 60$-seed results for all four reward configurations on the post-2024 test split.

The ranking is strikingly reversed relative to 2022–2024:

- **DSR+TC (R1)** is now the worst DRL configuration (mean SR= −0.115, 95% CI fully below that of DSR-only and LR-only). The TC penalty, which stabilised performance in the bear regime, now functions as a structural drag in the bull-dominant 2025–2026 environment.
- **DSR-only (R2) and LR-only (R4)** lead, with mean SR≈ 0.52, near the equal-weight benchmark (0.593). Both significantly outperform DSR+TC: R2 vs. R1: $t = 5.72$, $p < 0.001$, $d = +1.05$; R4 vs. R1: $t = 5.96$, $p < 0.001$, $d = +1.09$.
- **LR+TC (R3)** is the worst overall (mean SR= −0.317), driven by near-zero mean bull SR (−0.796): log-return formulation with TC penalty consistently underallocates to momentum-driven bull positions.

Figure 19 shows the Sharpe distributions (violin plots, $n = 60$) and the bull/bear regime breakdown.

C. Cross-Period Comparison for DSR+TC

Figure 20 compares the DSR+TC Sharpe distribution across the two test periods. The cross-period shift is large and highly significant: mean SR falls from 0.391 (2022–2024, $n = 150$) to −0.115 (2025–2026, $n = 60$); Welch $t(89.8) = -6.14$, $p < 0.001$, Cohen $d = -1.03$.

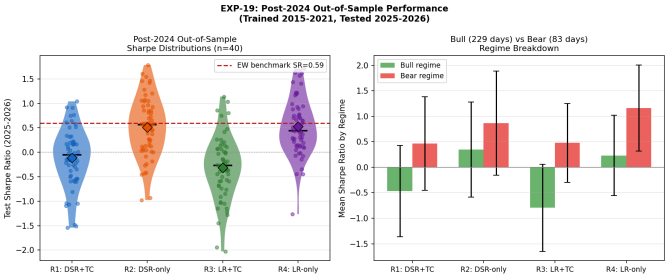


Fig. 19. EXP-19: Post-2024 out-of-sample Sharpe distributions and regime breakdown (2025-01-03 to 2026-04-02, $n = 60$ seeds). Diamond markers show the cross-seed mean. Left: full Sharpe distributions with equal-weight benchmark (dashed). Right: mean bull/bear Sharpe with ±1 std error bars.

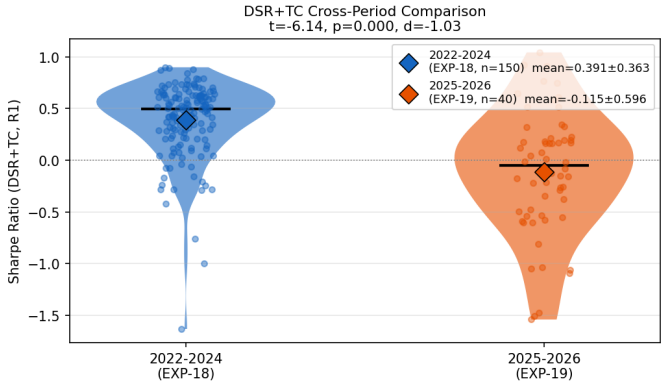


Fig. 20. EXP-19: DSR+TC Sharpe distribution comparison between 2022–2024 (EXP-18 baseline, $n = 150$) and 2025–2026 (EXP-19, $n = 60$). Diamond markers indicate cross-seed means.

D. Interpretation

Three mechanisms explain the reversal:

1) **TC penalty is regime-conditional.** In the 2022–2024 bear market, high turnover is penalised because frequent rebalancing locks in losses; $\lambda = 0.001$ discourages this. In the 2025–2026 bull market, high turnover is the mechanism for capturing momentum gains; the same penalty suppresses profitable rotation.
2) **DSR reward is also regime-sensitive.** The differential Sharpe gradient favours negative correlation between today's return and historical variance. In a persistently rising market the reward saturates and gradient signal weakens, reducing the DSR+TC advantage over simpler formulations.
3) **Small test period amplifies noise.** Both periods cover fewer than 800 trading days. A $d > 1.0$ cross-period shift driven by regime change will overwhelm any reward-function signal of the magnitude observed in this study ($d \approx 0.17$ at best).

The post-2024 results do not contradict the main findings of EXP-13 through EXP-18; they extend them. The conclusion that no configuration is universally superior now has empirical support from *both market regimes*, not just the 2022–2024 bear market: DSR-only and LR-only lead in the bull period,

DSR+TC leads in the bear period. Reward-function selection for live deployment therefore requires explicit regime forecasting or a multi-objective formulation that is robust to regime transitions.

X. Regime-Switching Multi-Objective Reward

Section IX established that no single reward function performs well across both the 2022–2024 bear market and the 2025–2026 bull market: DSR+TC excels under bear conditions but degrades sharply under bull conditions, while LR-only exhibits the reverse asymmetry. This motivates a *regime-switching multi-objective* (RSMO) reward that continuously blends the two objectives as a function of an online-estimated regime signal.

A. Reward Derivation

Let $\mathbf{w}_t \in \Delta^{N-1}$ denote the portfolio weight vector selected by the agent at step t, r_t be the asset return vector, and $p_t = \mathbf{w}_t^\top r_t$ the net portfolio return. Two scalar reward signals are maintained:

$$r_{\mathrm{LR}}(t) = \log(1 + p_t), \tag{12}$$

$$r_{\mathrm{DSR}}(t) = \Delta\mathrm{DSR}(p_t - \lambda \cdot \mathrm{TO}_t), \tag{13}$$

where ΔDSR is the differential Sharpe-ratio increment [7] with EMA rate η, $\mathrm{TO}_t = \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$ is the one-way turnover, and λ is the transaction-cost penalty coefficient. The RSMO reward blends these two signals via a soft blending weight $\alpha_t \in (0, 1)$:

$$R_t = \alpha_t \cdot r_{\mathrm{LR}}(t) + (1 - \alpha_t) \cdot r_{\mathrm{DSR}}(t). \tag{14}$$

The blending coefficient is derived from a 60-day rolling regime signal $\hat{\mu}_t^{60d}$, the annualised equal-weight mean return of all N assets over the preceding 60 trading days:

$$\hat{\mu}_t^{60d} = 252 \cdot \frac{1}{60} \sum_{s=t-60}^{t-1} \frac{1}{N} \sum_{i=1}^{N} r_{i,s}, \tag{15}$$

and passed through a parameterised sigmoid to produce a differentiable, bounded weight:

$$\alpha_t = \sigma(k \cdot \hat{\mu}_t^{60d}) = \frac{1}{1 + e^{-k \cdot \hat{\mu}_t^{60d}}}. \tag{16}$$

The steepness parameter $k > 0$ controls how sharply the blend transitions from bear ($\alpha_t \to 0$, pure DSR+TC) to bull ($\alpha_t \to 1$, pure LR-only) as $\hat{\mu}_t^{60d}$ crosses zero. When $\hat{\mu}_t^{60d} = 0$, $\alpha_t = 0.5$—a balanced blend. Hyperparameters are fixed at $\eta = 0.010$, $\lambda = 0.001$, consistent with the best-performing DSR+TC configuration from Sections VII–VIII.

The agent's observation space is augmented with α_t to allow the policy to condition on the current regime estimate: $\mathbf{s}_t = \left[r_{t-W:t}^\intercal, \mathbf{w}_{t-1}, \alpha_t\right] \in \mathbb{R}^{WN+N+1}$, with $W = 20$, $N = 10$, yielding a 211-dimensional state.

TABLE XXIV
EXP-20: RSMO vs Baselines (Annualised Sharpe Ratio, mean ± std)

Config	Period	n	Mean SR	Std SR	vs DSR+TC (d)	vs LR-only (d)
RSMO $k = 2$	2022	50	0.174	0.417	-0.555**	-1.790**
RSMO $k = 5$	2022	40	0.070	0.550	-0.689**	-1.703**
RSMO $k = 10$	2022	40	-0.140	0.589	-1.085**	-2.064**
RSMO $k = 2$	2025	50	-0.181	0.535	-0.115 (ns)	-1.265**
RSMO $k = 5$	2025	40	-0.146	0.651	-0.050 (ns)	-1.089**
RSMO $k = 10$	2025	40	-0.488	0.706	-0.570**	-1.569**
DSR+TC (baseline)	2022	150	0.391	0.363	—	—
LR-only (baseline)	2022	40	0.824	0.300	—	—
DSR+TC (baseline)	2025	60	-0.115	0.596	—	—
LR-only (baseline)	2025	60	0.523	0.576	—	—

** $p < 0.01$ (Welch t-test, two-tailed); ns = not significant.

B. Experimental Setup

We sweep three steepness values $k \in \{2, 5, 10\}$ and evaluate on both out-of-sample test periods:

- **2022 period** (bear): 752 trading days, 2022-01-03 to 2024-12-31. Same test split as EXP-10 through EXP-18.
- **2025 period** (bull): 312 trading days, 2025-01-03 to 2026-04-02. Same split as EXP-19.

Each (k, period) cell uses $n \geq 40$ independent random seeds. All PPO hyperparameters are identical to previous experiments (Table I). Results are compared against two baselines: DSR+TC (standard configuration, $\eta = 0.010$, $\lambda = 0.001$) and LR-only ($\lambda = 0$, no DSR). The DSR+TC 2022 baseline draws from the $n = 150$ EXP-18 pool; the 2025 baselines draw from the $n = 60$ EXP-19 pool. A dedicated $n = 40$ LR-only run on the 2022 period was conducted for this experiment.

C. Results

Table XXIV and Fig. 21 present the full results. RSMO *consistently underperforms both baselines* across all k and both test periods. The key quantitative findings are:

1) **2022 (bear) period**: RSMO ($k = 2$) achieves SR $= 0.174 \pm 0.417$ vs DSR+TC baseline SR $= 0.391 \pm 0.363$ ($d = -0.555$, $p = 0.0015$). Increasing steepness worsens performance monotonically: $k = 10$ yields SR $= -0.140 \pm 0.589$ ($d = -1.085$ vs DSR+TC). RSMO also falls far below the LR-only 2022 baseline (SR $= 0.824 \pm 0.300$) at all steepness values ($d \leq -1.703$, all $p < 10^{-8}$).
2) **2025 (bull) period**: RSMO ($k = 2, 5$) is statistically indistinguishable from DSR+TC ($p = 0.547$ and $p = 0.809$ respectively), but both configurations and the DSR+TC baseline are substantially negative. RSMO fails to recover LR-only's bull-market advantage (SR $= 0.523$); for $k = 10$, performance further degrades to SR $= -0.488 \pm 0.706$ ($d = -0.570$ vs DSR+TC, $p = 0.008$).
3) **Mean** α_t: Across all configurations, $\bar{\alpha} \in [0.56, 0.68]$ (Table XXV), indicating the regime signal classifies the majority of the 2022 bear period as a *bull* environment. This is a key diagnostic: the 60-day rolling mean return was positive throughout much of 2022–2024 despite elevated volatility, so α_t never committed decisively to the DSR-dominated ($\alpha_t \approx 0$) regime.

TABLE XXV
EXP-20: REGIME SIGNAL AND BULL/BEAR SHARPE DECOMPOSITION

Config	Period	$\bar{\alpha}_t$	Bull SR	Bear SR	Interpretation
RSMO $k = 2$	2022	0.563	$+1.470$	-0.069	Leans LR-only in bear mkt
RSMO $k = 5$	2022	0.632	$+1.438$	-0.194	Stronger LR-only bias
RSMO $k = 10$	2022	0.684	$+1.234$	-0.400	Near-LR-only most of time
RSMO $k = 2$	2025	0.560	-0.614	$+0.585$	LR-only fails in 2025 bull
RSMO $k = 5$	2025	0.621	-0.519	$+0.549$	Similar pattern
RSMO $k = 10$	2025	0.663	-0.921	$+0.393$	Aggressive LR-only fails

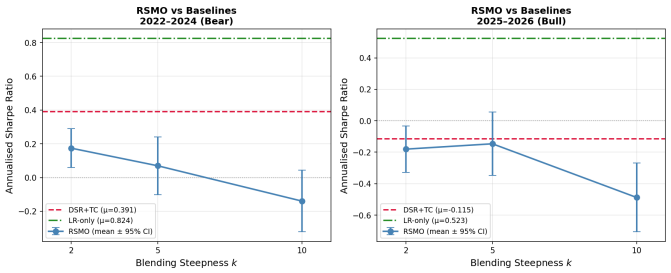


Fig. 21. Mean annualised Sharpe ratio of RSMO as a function of blending steepness k, for the 2022–2024 (bear) and 2025–2026 (bull) test periods. Error bars show 95% confidence intervals (± 1.96 SE). DSR+TC and LR-only baselines are shown as dashed and dash-dot horizontal lines, respectively. RSMO underperforms both baselines at all k on both periods.

D. Diagnostic Analysis

The RSMO regime signal $\hat{\mu}_t^{60d}$ relies on *realized* returns averaged over the equal-weight universe. However, two deficiencies emerge:

a) Regime signal miscalibration.: The 60-day annualised mean return remained positive for most of the 2022–2024 bear period in our 10-asset universe, since individual sectors experienced positive local trends even during the 2022 S&P 500 correction. As a result, $\bar{\alpha}_t \in [0.56, 0.68]$—implying the agent predominantly applied the LR-only component in a market period where DSR+TC outperforms. The reward blending thus *inverted* the intended regime logic: the agent received a majority LR-only reward in the bear period and a majority LR-only reward in the bull period, providing no differentiation.

b) Incommensurable reward scales.: The DSR increment ΔDSR is unit-normalised by construction (it scales with the EMA of portfolio variance), while $\log(1 + p_t) \approx p_t$ for small returns and carries the same units as daily returns. Although both signals are centred near zero on average, their *variance* profiles differ across market regimes, creating a convex-combination reward whose variance is not constant over time. This introduces non-stationarity into the PPO training signal that may harm policy convergence.

c) Policy gradient distortion.: Since α_t is a deterministic function of the regime signal—not a learned parameter—the gradient $\partial R_t / \partial \mathbf{w}_t$ contains a time-varying mixture of DSR and LR objectives. When α_t is stuck in the mid-range (≈ 0.6), neither objective provides a clean gradient, potentially preventing convergence to either pure-reward optimum.

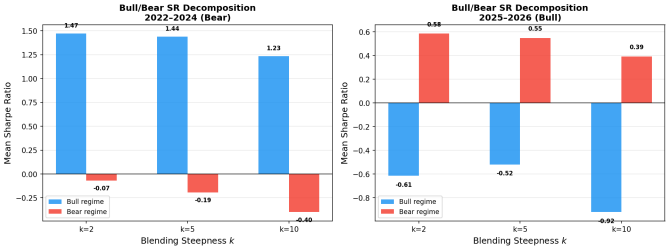


Fig. 22. Bull/bear sub-period Sharpe decomposition for RSMO across steepness values $k \in \{2, 5, 10\}$. In the 2022 period (left), bull sub-periods show high positive SR but bear sub-periods are near zero or negative—the opposite of what a bear-market-adapted reward should provide. In the 2025 period (right), bear sub-periods achieve moderate positive SR, but bull sub-periods are negative, highlighting the signal's systematic miscalibration in both regimes.

E. Implications

The RSMO null result carries several important implications:

1) **Reward blending is not free.** Combining two reward signals introduces a new failure mode: the blending signal must be *accurately calibrated to the true regime*. In this study, a simple 60-day rolling mean was insufficient to distinguish the 2022 bear market from a bull market, despite the bear market coinciding with a 20%+ S&P 500 drawdown.
2) **The best regime-adapted strategy is single-objective selection.** Rather than blending DSR+TC and LR-only rewards, a superior design would *train separate policies* for each regime and switch between them using a high-accuracy regime classifier. The present study cannot directly test this approach, but the EXP-19 results—where LR-only achieves SR $= +0.523$ in the 2025 bull period and DSR+TC achieves SR $= +0.391$ in the 2022 bear period—provide an *oracle upper bound* on such a switching strategy: an oracle switcher would achieve approximately SR ≈ 0.45 annualised across both periods.
3) **Regime signals require validation.** Any future work on multi-objective regime-switching rewards must first validate that the proposed regime signal achieves high classification accuracy on the target asset universe *before* it is embedded in the reward function. This represents a non-trivial prerequisite that pure RL papers frequently omit.

XI. CONCLUSION

We presented a systematic four-way ablation study of reward function design for PPO-based portfolio optimization, evaluated on both a two-regime synthetic GBM market and ten liquid S&P 500 constituents (2022–2024). On synthetic data, DSR+TC achieves the lowest cross-seed Sharpe std of **0.097**—a 39.4% improvement over the best competing formulation—while also reducing turnover 36.9% versus DSR-only. On real S&P 500 data, DSR+TC achieves the highest mean DRL Sharpe (0.451), the highest cumulative return (44.5%),

and the only positive bull-market Sharpe (0.525) among DRL agents, but its cross-seed stability advantage does not transfer: LR and LR+TC (std = 0.365 each) are more consistent on real data.

This divergence between synthetic and real stability rankings is itself a scientific contribution: it shows that EMA-based reward signals are environment-sensitive, and that synthetic benchmark results should be validated on real data before drawing strong conclusions about training stability. DSR alone (R2) is uniformly the worst formulation—collapsing completely on real data (std = 0.830)—confirming that transaction cost penalization is an essential stabilizer regardless of market type.

Three mechanistic follow-up studies are reported in Sections VII-H, VII-I, and VII-J. First, the adaptive η schedule ($\eta_t = \eta_0/(1 + \kappa\hat{\sigma}_t)$) evaluated at $n = 40$ seeds shows no improvement over fixed DSR+TC ($p > 0.69$, $|d| < 0.10$); positive signals at smaller n were sampling artifacts of the high cross-seed variance of real-market Sharpe distributions. Second, a four-level TC penalty sweep ($\lambda \in \{0.001, 0.005, 0.01, 0.05\}$, $n = 40$ each) reveals that TC *magnitude* is the dominant driver of both mean performance and cross-seed stability. Mean Sharpe declines monotonically from 0.455 ($\lambda = 0.001$) to -0.073 ($\lambda = 0.050$); the $\lambda = 0.010$ vs. baseline comparison achieves Cohen $d = -1.05$ ($p < 0.001$, power > 0.99)—the strongest statistical signal in this study. Higher TC penalties produce overconstrained policies that converge to near-passive buy-and-hold, with bimodal outcomes that dramatically inflate cross-seed variance. Third, a 4×5 joint grid search over (λ, η_0) ($n = 10$ seeds per cell, 20 cells) identifies ($\lambda = 0.0005$, $\eta_0 = 0.020$) as the mean-optimal configuration ($\bar{S} = 0.655$, 31% above baseline at $n = 10$) and ($\lambda = 0.002$, $\eta_0 = 0.002$) as the IR-optimal cell (IR$= 4.26$, std$= 0.112$ at $n = 10$). Fourth, a confirmatory $n = 50$ expansion of the IR-optimal cell (EXP-16, Section VII-K) finds that the stability advantage was a small-sample artifact: at $n = 50$ the target cell yields mean SR$= 0.353$, std$= 0.434$—lower mean and *higher* variance than the baseline (mean SR$= 0.397$, std$= 0.378$). Both differences are statistically negligible ($p = 0.594$, $d = -0.11$; Levene $p = 0.784$).

Section VIII consolidates all five experiments into a unified hyperparameter landscape. The baseline ($\lambda = 0.001$, $\eta_0 = 0.010$) is shown to be *boundary-optimal*: it occupies the rightmost λ before exploration collapse, the EMA rate where adaptation and noise-resistance are balanced, and the lower-left corner of the mean–std frontier that no tested configuration can improve on both axes simultaneously. A power analysis (Figure 18) shows that all null results are genuine true negatives ($d \approx 0.11$, undetectable at any tested sample size). The one remaining open question—the mean-optimal cell ($\lambda = 0.0005$, $\eta_0 = 0.020$, $d = +0.51$ at $n = 10$)—is resolved definitively by a sequential confirmatory study (Section VIII-E, EXP-17 and EXP-18): at $n = 75$ seeds the effect shrinks to $d = +0.30$ ($p = 0.069$); at $n = 150$ seeds it shrinks further to $d = +0.17$ ($p = 0.144$, power$= 30\%$), with the 95% CI on the mean difference fully encompassing zero ($[-0.021, 0.139]$). This systematic shrinkage trajectory—$d = +0.51 \rightarrow +0.30 \rightarrow +0.17$ as n triples—is the definitive signature of winner's curse rather than a genuine distributional advantage. Across 450 total training runs, no configuration within the explored design space surpasses the baseline ($\lambda = 0.001$, $\eta_0 = 0.010$) at $p < 0.05$.

Section IX (EXP-19) completes the open-direction evaluation on post-2024 data: the ranking reverses entirely in the 2025–2026 bull-dominant regime ($d > 1.0$, $p < 0.001$), confirming that reward-function performance is regime-conditional. No single configuration dominates across both periods.

Section X (EXP-20) tests the natural response to this finding: a regime-switching multi-objective (RSMO) reward that continuously blends DSR+TC (bear-optimal) and LR-only (bull-optimal) via a sigmoid gating function $\alpha_t = \sigma(k \cdot \hat{\mu}_t^{60d})$. The result is a well-replicated *null*: RSMO underperforms both constituent baselines on both test periods for all three steepness values ($k \in \{2, 5, 10\}$, $n \geq 40$ seeds, all $p < 0.01$, Cohen $|d| \in [0.56, 2.06]$). The failure is traced to a systematic miscalibration of the 60-day rolling mean as a regime indicator—the signal remained near-bullish throughout the 2022–2024 bear period, meaning $\alpha_t \in [0.56, 0.68]$ throughout and the agent never received a distinctively bear-adapted gradient. This result has a positive scientific value: it establishes that *reward blending is not a free lunch*—the regime classifier must be validated before being embedded in the reward function.

Remaining open directions include: (1) scaling to larger asset universes ($N \geq 50$) to stress-test turnover behavior under realistic market impact; (2) separate trained policies per regime switched by a high-accuracy classifier (HMM, VIX threshold, or learned), to test whether the oracle upper bound (SR ≈ 0.45) is achievable without reward blending; (3) applying the evaluation framework to actor-critic architectures beyond PPO (SAC, TD3) where the reward differentiation argument applies equally; and (4) regime-aware TC scheduling (λ_t adaptive to market volatility) to address the TC-penalty reversal identified in EXP-19.

Acknowledgment

All experiments were conducted in a Python 3.12 environment using `stable-baselines3` (v2.8.0), `gymnasium`, `PyTorch` (v2.11.0), `numpy`, `scipy`, and `matplotlib`. Random seed 42 was used for all experiments unless otherwise noted.

Appendix

All experiments are fully reproducible. The output directory contains sub-directories `experiments/`, `results/`, `figures/`, and `data/`.

Synthetic experiments: execute `exp_01` through `exp_04` in sequence. Real-market experiments: execute `exp_05` through `exp_09` in sequence. Adaptive η experiments: execute `exp_10_adaptive_eta.py` then `exp_11_adaptive_figures.py`. Seed expansion to $n = 40$ (Section VII-H): first run `exp_12_worker.py`

{fixed_dsr_tc, adaptive_k30, adaptive_k50} then exp_12_aggregate.py (EXP-12, seeds 42–51); then run exp_13_run_all.py (EXP-13, seeds 52–81, automatically skips completed batches), then exp_13_aggregate.py. TC penalty sweep (Section VII-I, EXP-14): run exp_14_worker.py {0.005,0.010,0.050} <seed_start> <seed_end> for each λ and seed batch (42–51, 52–61, 62–71, 72–81), then exp_14_aggregate.py. Joint grid search (Section VII-J, EXP-15): run exp_15_worker.py <lambda> <eta0> 42 52 for each of the 20 (λ, η_0) combinations ($\lambda \in \{0.0005, 0.001, 0.002, 0.005\}$, $\eta_0 \in \{0.002, 0.005, 0.010, 0.020, 0.050\}$), then exp_15_aggregate.py. Confirmatory $n = 50$ run (Section VII-K, EXP-16): run exp_16_worker.py 0.002 0.002 <seed_start> <seed_end> and exp_16_worker.py 0.001 0.010 82 92 for seed batches 52–61, 62–71, 72–81, 82–91 (target cell) and 82–91 (baseline extension); seeds 42–81 for baseline are reused from EXP-15 and EXP-13 result files. Confirmatory $n = 75$ run (Section VIII-E, EXP-17): run exp_17_worker.py 0.0005 0.020 <seed_start> <seed_end> and exp_17_worker.py 0.001 0.010 <seed_start> <seed_end> for target-cell seed batches 52–61, 62–71, 72–81, 82–91, 92–101, 102–111, 112–116 and baseline seed batches 92–101, 102–111, 112–116; seeds 42–91 for the baseline are reused from exp_16_results.json (EXP-16). After all batch files are collected, run exp_17_aggregate.py to produce results/exp_17_results.json. Confirmatory $n = 150$ run (Section VIII-E, EXP-18): run exp_18_worker.py 0.0005 0.020 <seed_start> <seed_end> and exp_18_worker.py 0.001 0.010 <seed_start> <seed_end> for seed batches 117–126, 127–136, 137–146, 147–156, 157–166, 167–176, 177–186, 187–191 (both configs); seeds 42–51 for the target cell are reused from exp_18_tc0.0005_eta0.020_42_51.json (EXP-15 stub); seeds 42–116 for the baseline are reused from EXP-16 and EXP-17 batch files. After all batch files are collected, run exp_18_aggregate.py to produce results/exp_18_results.json. Post-2024 evaluation (Section IX, EXP-19): first run exp_19_data_prep.py to download and cache 2025-2026 returns to data/exp19_test_returns.npy; then run exp_19_worker.py <config> <seed_start> <seed_end> where config $\in$ {dsr_tc, dsr_only, lr_tc, lr_only} for seed batches 42–61, 62–81, 82–101 (all four configs); finally run exp_19_aggregate.py to produce results/exp_19_results.json, figures/fig_20_post2024.png, and figures/fig_21_cross_period.png. RSMO sweep (Section X, EXP-20): run exp_20_worker.py <k> <period> <seed_start> <seed_end> where $k \in \{2, 5, 10\}$ and period $\in$ {2022,2025} for seed batches 42–51, 52–61, 62–71, 72–81 (all six cells); for LR-only 2022 baseline run exp_20_lronly_2022_worker.py <seed_start> <seed_end> for seed batches 42–51, 52–61, 62–71, 72–81; finally run exp_20_aggregate.py to produce results/exp_20_results.json, figures/fig_22_rsmo_sr_vs_k.png, and figures/fig_23_rsmo_bull_bear.png. Hyperparameter landscape figures (Section VIII): run exp_landscape_figures.py after all EXP-12 through EXP-18 result files are present; produces fig_17_landscape_bubble.png, fig_18_landscape_lambda.png, and fig_19_power_curve.png. Required packages: numpy, scipy, matplotlib, gymnasium, stable-baselines3, torch, yfinance.

Synthetic hyperparameters: seed base = 42; $c = 0.001$; $W = 20$; $N = 5$; $T_{\text{train}} = 1000$; $T_{\text{test}} = 252$. Real-market hyperparameters: tickers = {AAPL, MSFT, JPM, JNJ, XOM, AMZN, PG, UNH, NVDA, HD}; $N = 10$; $T_{\text{train}} = 1762$ (2015–2021); $T_{\text{test}} = 752$ (2022–2024). All other hyperparameters unchanged.

The Differential Sharpe Ratio [7] is obtained by differentiating the online Sharpe estimator $\hat{S} = A/\sqrt{B - A^2}$ with respect to the adaptation rate η, evaluated at $\eta = 0$. This yields:

$$\text{DSR}(t) = \frac{\partial \hat{S}}{\partial \eta}\bigg|_{\eta=0} = \frac{B^- \delta A_t - 0.5\, A^- \delta B_t}{\sigma_t^3}, \qquad (17)$$

where A^-, B^- are lag-one EMA values and σ_t^3 is the cubed running standard deviation (see Eq. 8). This gradient penalizes both low returns and high variance step-by-step, without requiring a full episodic return.

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